

More with Less: The Impact of Mandatory Overtime on Police Wellness and Productivity

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Abstract

Can mandatory overtime effectively serve as a buffer against labor shortages, or does the cost of increased exhaustion and diminished productivity outweigh the benefits? We address this critical question by analyzing a mandatory overtime policy implemented by the Chicago Police Department (CPD) in 2022-2023. This policy created both cross-sectional and temporal variation in overtime assignments, which we leverage in a “stacked” difference-in-differences design. Using a comprehensive dataset of daily officer-level records, we evaluate the effects of assignment to mandatory overtime on officer behavior, wellness, and on-the-job performance. Our primary findings indicate that being assigned one additional shift does not subsequently increase officer injuries, sick leave, or overall absence rates. We also do not find evidence that the policy affected key enforcement outcomes, including arrests, calls for service, and the use of force. Our analysis is robust; for several outcomes, such as officer attendance and on-the-job injuries as well as response times, our findings are precise enough to rule out even modest changes. For rare events like arrests and the use of force, our estimates are congruent but less precise. These results collectively suggest that the limited and targeted use of mandatory overtime to manage labor shortages can be done in a manner that poses little risk to officer wellness or productivity.

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1 Introduction

Across the United States, public sector institutions— including schools, hospitals, and police departments— are grappling with severe staffing shortages and challenging work environments (Doan et al., 2024, Maryland Office of Legislative Audits, 2024, Pandemic Response Accountability Committee, 2023). These challenges have been exacerbated by the COVID-19 pandemic and the retirement of an aging workforce. This context is particularly salient for law enforcement agencies, where high vacancy rates have led to concerns about increased workloads and declining officer morale (City of Chicago Office of Inspector General, 2022, Police Executive Research Forum, 2023), further complicated by shifts in public sentiment and increasingly tight budgets (International Association of Chiefs of Police, 2020, National Conference of State Legislatures, 2021). The widespread perception, supported by a growing body of literature, is that excessive workloads negatively affect workers. However, the effects of modest and distributed increases in workload— common strategies for managing staffing shortages in law enforcement— are less well understood.

Our paper fills this gap by directly examining how short-term, mandatory overtime, particularly that resulting from the cancellation of a scheduled day off, affects officer behavior, wellness, and on-the-job performance. Our study focuses on the Chicago Police Department (CPD), which in 2022 introduced a new system of cyclical, pre-scheduled cancellations of Regular Days Off (RDOs) to address significant staffing shortages and meet mandatory staffing requirements. While Chicago had been increasingly leveraging mandatory overtime since the COVID-19 pandemic, we focus on a specific mandatory overtime policy associated with CPD’s use of a tiered deployment model, beginning in February 2022 and ending in early 2024. This policy involved canceling the first of an officer’s two scheduled RDOs once every six weeks, resulting in a single extended workday followed by a day of rest. Crucially, this cyclical system was distinct from traditional RDO cancellations because it was determined idiosyncratically and in advance, not tied to specific events, with each cancellation affecting hundreds of officers simultaneously. Thus, the broad policy and the nuances of the rotation schedule generate plausibly exogenous variation in officer workloads. This variation, unrelated to individual officer behavior or short-term operational needs, allows for an identification strategy which allows us to isolate the short-run causal effect of the policy.

To estimate the effects of these cancellations, we use linked personnel, dispatch, and scheduling data from February of 2022 to December of 2023. We identify RDO cancellation events using department-wide “tiered deployment” schedules which include a matrix of cancellations and construct a balanced panel of officers

with rotating work-rest patterns. Our empirical strategy considers officers with otherwise identical work schedules, comparing officers who have one RDO canceled to others assigned to their usual work schedule. We compare outcomes for officers in the first group to those from the second in the days before and after an RDO cancellation. In particular, we implement a so-called “stacked” event study design with officer, date, and assignment fixed-effects. Our empirical design mitigates concerns about bias related to staggered rollout by excluding early-late comparisons that arise in a two-way fixed-effects specification (de Chaisemartin and D’Haultfoeuille, 2023, Deshpande and Li, 2019, Goodman-Bacon, 2021, Wing et al., 2023). This approach allows us to isolate the causal effect of short-term, externally imposed increases in workload on officer behavior, performance, and wellbeing. This setting offers several advantages: First, it isolates plausibly exogenous, short-run variation in workload, circumventing common confounders related to officer selection for overtime or high-activity shifts. Second, it enables us to track outcomes with high temporal precision. Third, it offers insight into a common policy used by public safety organizations and other frontline agencies to manage chronic staffing shortages.

Leveraging our causal identification strategy, we demonstrate that assignment to a canceled RDO significantly increases an officer’s likelihood of working that day. This confirms broad compliance with the policy and validates our use of an intent-to-treat analysis, given that actual compliance may be endogenous. We find that these RDO cancellations have no meaningful impact on officer productivity or wellness. Our estimates show that officers respond to the same number of calls for service and arrive just as quickly in the days following a cancellation, which we characterize as precise zeros. We also find no change in discretionary enforcement activity, including stops, arrests, and use of force, though these latter estimates are less precise. We also find that officers are no more likely to experience an injury or call in sick in the days after a canceled RDO. These results are robust to a range of modeling choices, and our heterogeneity tests reveal no significant differences in treatment effects by officer demographics, enforcement patterns, or prior exposure to RDO cancellations.

Taken together, this evidence suggests that short-term, externally imposed increases in workload have minimal impact on how officers behave or feel once they return to their standard work schedule. However, the data suggest this null result may be driven by the specific nature of the additional work. Despite institutional documents suggesting that officers are engaged in typical duties during these additional shifts, our findings show that officers working a canceled RDO are significantly less likely to respond to calls for service or make arrests. This discrepancy is likely explained by concurrent enforcement initiatives active during

our treatment period that prioritized high-visibility deterrence. Officers assigned to these details were tasked primarily with maintaining a static presence rather than engaging in proactive patrol or responding to calls for service. Therefore, while the treatment represents a significant imposition on officer leisure time through the mandatory loss of an RDO, the shifts themselves appear to consist of lower-intensity duties compared to a standard workday.

This study contributes to three strands of literature: burnout, overtime and police performance, and public safety management. The existing literature on high-burnout occupations documents the long-term consequences of cumulative fatigue and extreme stressors on worker productivity and turnover (Blackburn et al., 2023, Cho et al., 2024). Research on healthcare workers and other frontline occupations consistently shows that long hours and stressful conditions can reduce productivity, impair judgment, and increase errors (Blackburn et al., 2023, Lu and Lu, 2017, Nekoei et al., 2024, Sachiko and Isamu, 2016). A related literature focuses on short-run increases in hours or consecutive days worked, particularly in policing where these policies are prevalent. Descriptive evidence from the NYPD finds that overtime correlates with higher use of force, more complaints, and officer injuries (New York City Department of Investigation, 2023). While burnout reflects chronic strain, even temporary workload increases can mimic early symptoms of burnout and shape officer behavior. Complementary evidence comes from experimental settings: fatigued night-shift officers perform worse in simulated scenarios (James, 2018).

Our project is most closely related to concurrent work by Ferrazares (2025), who also uses data from the Chicago Police Department. Their study leverages a well-accepted identification strategy that exploits idiosyncratic variation in scheduling assignments (Ba et al., 2021, Gudgeon et al., 2024, Holz et al., 2023). Rather than examining a discrete policy, Ferrazares investigates the cumulative effects of consecutive workdays, finding that each additional day worked increases the risk of both use of force and officer injury—a pattern consistent with fatigue and burnout. Our study differs in two important ways. First, we focus on the impact of a specific policy: a single mandatory overtime day imposed on officers, followed by a guaranteed period of rest. Second, by focusing on discrete scheduling shocks rather than cumulative exposure, our design allows us to directly analyze the immediate, short-run effects of overtime as a management tool, rather than its long-term consequences.

Our findings align with a strain of the management literature that identifies contexts in which marginal increases in workload can be absorbed without measurable harm. In healthcare, Kuntz et al. (2015) document a “tipping point” in hospital mortality, showing that hospitals can absorb marginal patients until

very high occupancy rates, while Kc and Terwiesch (2009) find that hospitals can increase service rates under rising workloads. Comparable evidence in policing comes from Cho et al. (2024), who show that enforcement activity falls after a line-of-duty death without increasing crime, suggesting that adjustments in policing intensity can be absorbed without jeopardizing public safety. Taken together, the existing literature on the cumulative effects of burnout and this strain of management literature suggests that fatigue-related harms may arise over the long run but only when exposure is sustained without adequate recovery. Our study contributes to this literature by analyzing a distinct scheduling margin—a single mandatory overtime day followed by rest—and finding no impact on short-term productivity and well being.

Finally, a separate and related literature highlights mechanisms that may mitigate the negative impact of excessive work and interventions that may reduce their deleterious effects. For example, Chalfin and Goncalves (2020) demonstrate that police officers’ pro-social motivations may influence how they respond to additional demands which can mediate the relationship between workload intensity and outcomes. Several studies suggest that interventions such as peer support and compressed work schedules have the potential to mitigate burnout. Linos et al. (2022) show that dispatcher peer-support programs reduce burnout, and Fan et al. (2025) find a significant reduction in burnout from a four-day work week. In the short run, our study suggests that these interventions may not be necessary if mandatory overtime is strategically scheduled such that the affected day is followed by a guaranteed period of rest.

Our findings have practical implications for police leaders navigating chronic staffing shortages. While public debate often focuses on sweeping reforms or major recruitment drives, departments typically manage understaffing through marginal adjustments, such as overtime and extended shifts. The Chicago Police Department’s systematic and cyclical day-off cancellations exemplify such incremental responses, implemented without the confounding influence of emergent operational needs. This setting reflects the broader reality of high overtime in frontline agencies. For instance, in 2021, Chicago officers already averaged 11 overtime hours weekly on top of a 54-hour schedule. Against this backdrop, our results suggest that modest increases in mandatory overtime do not generate measurable short-term harms—and that equally modest reductions may not yield substantial improvements. While larger structural changes could still affect burnout or morale, our evidence indicates that the marginal adjustments most departments rely on are unlikely to meaningfully shift outcomes. These findings may also extend to other public-sector workforces where overtime is a routine stopgap, particularly frontline workers.

2 Institutional Context, Data, and Sample Construction

2.1 Institutional Context

The Chicago Police Department (CPD) provides a useful setting to study the impacts of mandatory overtime. Like many jurisdictions in the United States, CPD experienced gradual declines in the number of sworn officers over the last two decades. These declines became especially acute between 2019 and 2022, when retirements and resignations among sworn Chicago police officers nearly doubled, producing a 10% reduction in patrol officers. Although it is difficult to pinpoint the specific causes of these departures, they coincided with the upheaval of the COVID-19 pandemic and a wave of nationwide protests against excessive use of force by police.

In response to these staffing shortages and the protests that followed the killing of George Floyd, CPD began canceling officers' scheduled days off in May 2020. The goal was to ensure sufficient staffing during periods of heightened crime and disorder. Over the subsequent three years, CPD continued to rely on day-off cancellations, effectively mandating overtime across the force. In February 2022, the department formalized this practice by instituting pre-determined cancellations that cycled through groups of officers. Importantly, officers had little choice over when these mandatory days off occurred.¹ Concerns about the toll on staff soon surfaced. A 2022 audit revealed that some officers worked as many as 12 consecutive days with shifts as long as 12 hours in April and May of that year (City of Chicago Office of Inspector General, 2022). The policy remained in place until October 2023, when newly appointed Superintendent Larry Snelling announced a general prohibition on mandatory overtime, limiting it only to holidays and special events beginning in 2024 (Sisk, 2023). While this termination marked the end of the scheduling mandate, it was also driven by a re-evaluation of how these additional hours were being utilized in practice.

A notable portion of these mandatory overtime hours were allocated to a high-visibility enforcement strategy colloquially known as "Operation Scarecrow." Officially aimed at deterring crime through static presence, this initiative required officers to man fixed posts in marked vehicles with emergency lights activated, particularly within the downtown business districts (Masterson, 2024). The scope of this deployment was significant; in 2022 alone, the department utilized over one million overtime hours to staff these specific shifts. However, the nature of

¹Additional mandatory overtime was also assigned during special events and holidays independently of this policy.

this work differed largely from standard patrol duties. Rather than responding to calls for service or engaging in active policing, officers on these assignments were effectively functioning as visual deterrents. The initiative drew criticism for treating officers as “props” rather than law enforcement agents, a factor cited by Superintendent Snelling when he eventually dismantled the program in late 2023 to prioritize officer morale and effective deployment (NBC Chicago, 2023, Schuba and Main, 2024). This context is crucial for our analysis, as it suggests that a subset of the mandatory shifts imposed during our study period involved low-intensity, passive monitoring rather than the high-stress demands of typical patrol work. Ideally, we would distinguish between mandatory shifts used for standard patrol and those used for static deterrence; however, administrative data limitations prevent us from disentangling these specific assignments. Consequently, our estimates should be interpreted as the intent-to-treat effect of the mandatory overtime policy, capturing a weighted average of standard patrol duties and these lower-intensity deployments.

The aggregate impact of these initiatives is readily apparent in the raw administrative data. As shown in Figure 1, the cumulative number of days when officers claimed overtime grew by 59% from a mean of 175 thousand between 2014 and 2019 to nearly 250 thousand in 2023. Similarly, the cumulative number of overtime hours grew by 56% over the same period. Between 2019 and 2020, involuntary overtime days and hours increased by 172% and 134%, respectively. The number of both days and hours of involuntary overtime remained at elevated levels through 2023. From 2021 to 2023, the rate of voluntary overtime in terms of days and hours grew by more than a factor of four. These figures highlight the sheer scale of overtime and CPD’s growing reliance on both voluntary and involuntary overtime, a broader context that extends beyond our study’s focus on a specific mandatory policy.

Understanding CPD’s work assignment system is critical for interpreting the overtime policy. Beginning long before the implementation of the mandatory overtime policy, officers at CPD have been assigned to a “day-off group” (DOG) that follows a repeating six-day cycle. For patrol officers, this would typically consist of four consecutive work days followed by two consecutive rest days. DOG assignments go into effect in January of each year and are determined through a seniority-based bidding process in the preceding fall. Although officers can and do change DOGs throughout the year— we observe switching in nearly 40% of officer-years— these changes remain subject to managerial discretion. In addition to a DOG assignment, patrol officers at CPD are also assigned to a particular geographic unit. The city is divided into five patrol areas, which contain 22 districts and 277 beats. While other empirical studies using CPD data have

leveraged idiosyncratic variation from January DOG assignments, our strategy uses a distinct source of identifying variation.² Specifically, our empirical design relies on the tiered-deployment RDO cancellation matrix, which varies over time by both assigned DOG and patrol area.

As mentioned, the mandatory overtime from the tiered-deployment policy was operationalized through a "cancellation matrix"—a set of pre-determined calendars that revoked RDOs for all officers assigned to a specific combination of DOG and patrol area. Each cycle began in a particular area, with cancellations cycling sequentially through each DOG assignment within that area. Subsequent cycles repeated this process in a different area. This rotating system meant that the timing of mandatory overtime was driven by a departmental schedule, not by officer choice. For there to be endogenous selection in our setting, one would have to imagine that officers obtain manager approval to move between DOG assignments and assigned areas based on the timing of the canceled RDO on the matrix. We believe, and our event study estimates corroborate, that this is unrealistic and that the policy created plausibly exogenous variation in which officers were required to work on their RDO.

2.2 Data

Our analysis utilizes administrative data primarily obtained from Freedom of Information Act (FOIA) requests filed with the CPD. Officer attendance data is sourced from the CPD’s Attendance and Assignment (A&A) database, which is effectively "punch card" data that records the start and end of each shift and includes a detailed paycode. With nearly a hundred options, these paycodes are highly specific, documenting whether an officer was on various types of regular or overtime duty. The data also capture the officer’s shift, unit, and beat for a given day, and notes the reason for any absence, such as paid time off, a workplace injury, or an RDO. While the A&A database tracks daily work and absences, we separately use a formal assignments database to observe each officer’s official DOG and area assignment over time. We measure treatment by combining these formal assignments with the RDO cancellation matrix. An officer is considered treated by the policy if their DOG-area combination is scheduled to have its RDO canceled. Since supervisors can grant exemptions, resulting in imperfect compliance, we focus on intent-to-treat effects. We define an officer as treated if they are scheduled to work on their RDO, regardless of whether they ultimately do.

²Recent research has also used CPD’s DOG system to study how officer race, gender, and experience shape performance (Ba et al., 2021, Gudgeon et al., 2024).

We study outcomes in three domains: officer wellness, productivity, and enforcement. Each of these measures captures a distinct channel through which mandatory overtime might affect officers and the public. We measure wellness using various reasons for an absence reported in the A&A data. Specifically, we construct daily indicators for whether an officer (1) was absent due to an injury sustained on the job, or (2) called in sick to work. We also consider attendance and overtime in the week following a canceled RDO as complementary measures of wellness. Together, these outcomes capture whether mandatory overtime increases short-term illness or injury risk as well as whether officers make their own workload adjustments to mitigate the immediate effects of the policy.

We measure productivity by counting the number of incidents involving each officer, either through responding to a call for service (CFS) or by initiating an investigatory stop. CFS data capture incidents for which a 911 dispatcher requested and received a police response, and thus reflect responses to public demand for police assistance. Investigatory stop data capture officer-initiated encounters in which an officer temporarily detains an individual suspected of criminal activity. We obtained universe files for both incident types from CPD and construct daily officer-level counts of CFS responses and investigatory stops.³ Since both activities involve some degree of discretion (less in CFS, more in investigatory stops), mandatory overtime may reduce engagement if overburdened officers choose to shirk. We also measure the time between a dispatcher’s receipt of the CFS and officer’s arrival on the scene, included as a supplementary outcome in Figure C.10a.

Our third domain—enforcement activity—is closely related to productivity but conceptually distinct. Whereas our productivity measures capture selection into an incident, our enforcement measures capture the intensity of those interactions via the decision to make an arrest or use force. The arrest data provide the universe of CPD arrests with information on charges and all officers involved (primary, secondary, or assisting). Because arrests are relatively rare at the officer-day level, we define an arrest indicator equal to one if an officer contributed to any arrest that day. Use of force data identifies officers who reported using physical force. CPD requires officers to document force used against “actively resisting” civilians, resulting in reported injury, or involving civilians who threatened, obstructed, or assaulted an officer (Chicago Police Department, 2023).⁴

³We top code officer daily CFS volume at the 99th percentile or 21 calls in a single day.

⁴In 2022–23, 79% of reports indicated use of control tactics (e.g., holds, restraints); 61.7% reported bodily force without weapons (e.g., punches, tackles); and 4.4% reported use of a weapon (e.g., tasers, batons, firearms).

2.3 Sample Construction

While the policy was active from February 2022 to December 2023, we focus our analysis on the period between May 31, 2022, and October 20, 2023, excluding a three month period between January 3 and April 4, 2023. We limit our study to this specific time frame because the cyclic RDO cancellations were most consistently documented and implemented during these periods.⁵ Of the total 13,601 officers (93% of all officers) in the roster that we are able to link across all of our data, we restrict the sample to 7,927 patrol officers because their schedules are most directly impacted by the policy. The majority of patrol officers were assigned to one of six DOGs (61–66). All of these DOG schedules followed a four-days-on to two-days-off rotation. These DOGs are sequentially offset by one day, ensuring consistent staffing throughout the week. We also restrict the sample to patrol officers assigned to the five main geographic areas (Areas 1–5). Citywide units, which operate across multiple areas, were excluded because they did not follow the same cancellation pattern.⁶

Cancellations fall on the first rest day of each DOG cycle. Complying officers therefore worked on that day, rested the next, and then restarted their DOG cycle. We use this structure to construct a series of “sub-experiments”. For each cancellation, we compare officers in the same DOG but assigned to different areas such that only some experienced the RDO cancellation. Our analytical sample follows officers for five days before and after each RDO cancellation, capturing the work cycles directly surrounding treatment. This window allows us to measure immediate changes in wellness (injuries, sick leave, attendance), productivity (calls for service and investigatory stops), and enforcement (arrests and use of force). For each sub-experiment, we construct balanced panels, excluding officers we do not observe in all of the data within the respective period. For control units, we additionally require that officers were not treated within an eleven-day window around the focal cancellation.⁷

Pooling across 288 observed cancellations yields a stacked sample covering roughly 48% of patrol officer-dates during the study period.⁸ In the analytical

⁵We omit the first and last months of the policy because of irregular and inconsistent implementation. We omit much of Q1 2023 because of a gap in the availability of cancellation matrices during this period.

⁶To minimize potential noise from short-term assignment changes, we impute DOG and area by officers’ modal assignment within each month.

⁷In particular, we omit officers treated in the work cycle (6 days) prior to the focal cancellation, or during the remainder of the work cycle beginning with the focal RDO (5 days). We provide results from longer- and alternative short-run windows in the Appendix.

⁸The full linked data cover 7,927 officers and 3.2 million officer-days. The stacked sample consists of 5,864 officers and 1.9 million officer-days. Demographics and baseline outcomes are nearly identical between the full and stacked samples, aside from a slightly higher share of Black officers (1.7 percentage points). Detailed balance tables are in Table 1.

sample, 34% of officer-dates fall are treated units and 64% are control units. As shown in Table 1, absences are somewhat common (on an average day, 4.4% of officers call out injured, and 6.1% sick, with additional officers taking paid time off. Officers average 2.4 responses to calls for service per day, initiate 0.06 investigatory stops, make 0.08 arrests, and use force on 0.17% of days. These outcomes provide the empirical foundation for assessing how mandatory overtime reshaped wellness, productivity, and enforcement.

3 Empirical Design

Our empirical design relies on the quasi-random assignment of officers to overtime introduced by RDO cancellations. The cyclical pattern of these cancellations, combined with DOG and area assignments for each officer, generates within-DOG and between-area variation. In effect, our design compares officers in the same DOG who would otherwise have identical schedules, but differ due to RDO cancellations in some areas. As described in Section 2.2, we consider each RDO cancellation as a single natural experiment induced by the overtime policy. By pooling data across sub-experiments in a stacked sample, our difference-in-differences design allows us to leverage all RDO cancellations we observe to estimate the effects of mandatory overtime on a range of officer-level outcomes. This design avoids problematic comparisons between early and later treated observations that can arise in a two-way fixed-effects specification in the presence of treatment heterogeneity (de Chaisemartin and D’Haultfoeuille, 2023).

To explore potential temporal dynamics in our outcomes and to assess pre-treatment trends across groups, we estimate event study models of the following form:

$$\begin{aligned}
Y_{iet} = & \alpha_1 D_{ie} + \sum_t \delta_t D_{ie} \times T_t \\
& + \alpha_2 DaysLastTreated_{iet} + \phi_i + \kappa_{ie}^{A \times d} + \kappa_{ie}^{D \times d} + \epsilon_{iet}
\end{aligned} \tag{1}$$

Here Y_{iet} reflects the outcome of interest for officer i in sub-experiment e at event time t . D_{ie} is officer i ’s treatment assignment in sub-experiment e – recall that this treatment assignment will vary within officers across sub-experiments as officers not called to work on a given RDO can be in the treated group in a future or past sub-experiment. We include a set of treatment by event time interactions $D_{ie} \times T_t$. We exclude the treatment-event time interaction for the day before a cancellation ($t = -1$), so the parameters are estimated relative to

the difference in treatment and control at event time $t = -1$. We also control for $DaysLastTreated_{iet}$, the number of days since an officer was last treated and include officer (ϕ_i), area - by - date ($\kappa_{ie}^{A \times d}$), and DOG -by- date ($\kappa_{ie}^{D \times d}$) fixed-effects.

The set of δ_t on the interactions of treatment and event time are our coefficients of interest. We consider officers who change DOG-area assignments within a given month to be non-compliers, and therefore interpret the δ_t as an intent-to-treat (ITT) estimate. While we include the canceled and non-canceled RDO in our event study estimates in order to show compliance, we omit these days from aggregated difference-in-difference estimates across the whole post period. Specifically, we also estimate an aggregated δ across the post-periods where we modify equation 1 to include only a single post-treatment indicator and interaction, which excludes RDOs.

Following Wing et al. (2023), we specify how our estimator aggregates treatment effects across sub-experiments. In particular, we apply observation-level weights so that our estimates, $\hat{\delta}_t$, correspond to weighted averages of sub-experiment ITTs, with weights equal to each sub-experiment's sample share, shown in 2.

$$\delta_t = \sum_{\Omega_e} ITT(e, t) \frac{N_e}{N} \quad (2)$$

N_e is the number of total observations in sub-experiment e , N is the size of the total stacked sample, and Ω_e is the universe of sub-experiments. To achieve this estimand, we weight treatment and control observations as follows in 3, where N^g denotes the number of observations in the group (treatment or control) to which the weights will apply:

$$\frac{N_e/N}{N_e^g/N^g} \quad (3)$$

Because treatment assignment is determined by the combination of an officer's area and DOG assignments, we cluster standard errors at the area-DOG level.⁹

⁹Alternative standard error assumptions like clustering on officer or twoway clustering by DOG and area yield qualitatively similar results.

3.1 Identifying Assumptions

Our empirical strategy enables us to avoid key sources of endogeneity in the relationship between overtime and worker performance. First, because overtime is normally voluntary, individuals who work extra hours might systematically differ from those who do not. We overcome this selection bias by examining mandatory overtime imposed through RDO cancellations; our intention-to-treat (ITT) estimates are robust to imperfect compliance. Second, the demand for police overtime often correlates with spikes in crime, potentially confounding results. We mitigate this by leveraging the CPD’s predetermined, sequential RDO cancellation pattern, which is determined months in advance and orthogonal to immediate crime trends or demand fluctuations. Third, we use a daily officer panel within a difference-in-differences framework, incorporating officer, area - by - date, and DOG- by - date fixed-effects, to account for non-random officer selection into specific assignments, areas, and initial level differences in working conditions. We previously noted that officers deviate from their January assignments and may endogenously select into their DOG or area assignments. However, our data show no evidence of officers being selected within their DOG by area assignments in response to the timing of the RDO cancellation schedule.

Our difference-in-differences identification strategy relies on several standard assumptions. The “no anticipation effects” assumption requires that officers do not change behavior in advance of a cancellation. Empirically, our event study plot in Figure 2b shows no evidence of anticipation effects for absences. In Appendix section C.1, we also show that officers do not differentially change DOG or area to evade treatment. Next, the parallel trends assumption requires that treatment and control group outcomes would evolve similarly without the canceled RDO. This is plausible given our observation of comparable pre-treatment trends for all outcomes; we present event study estimates for the four regular work days preceding treatment for each outcome in section 4. While we are powered to detect minor variations in pre-treatment event study estimates, these statistical violations are best characterized as idiosyncratic or small and do not amount to meaningful evidence of differential pre-trends that would invalidate our inference.

Furthermore, because CPD implemented cancellations sequentially and across the entire city, it is unlikely that our results are driven by localized shocks or unexpected events that happened to overlap with treatment. An alternative concern is that the broad scope of officers affected by cancellations allows our control groups to include officers who were treated in other sub-experiments. However, this does not introduce bias if treatment effects are short-lived – specifically, if they do not persist into subsequent work cycles after our evaluation window. This

is reasonable in our institutional setting, where mandatory overtime occurs no more than once a month and each work cycle with a canceled RDO is followed by two days off. In addition, our regressions explicitly control for the number of days since an officer was last treated. As a robustness check, we relax this assumption and allow treatment effects to last for two full work cycles instead of one. The results are highly similar under this alternative, as shown in section C.3 and section C.4.

4 Results

4.1 Treatment Compliance

In this paper, we estimate the effects of overtime on officer wellness, productivity, and enforcement outcomes using RDO cancellations as conditionally quasi-random variation in days worked. Because officers may switch assignments or take alternative leaves throughout the year, we neither expect nor observe perfect attendance (compliance with the policy) for the treated group or perfect absence for the control groups.¹⁰ However, we can clearly demonstrate the validity of the first stage for producing intent-to-treat estimates: on average, officers do work more when their RDO is canceled.

We begin by showing evidence of compliance in the raw data. Based on CPD operations calendars, we expect officers to work four consecutive days and then rest for two before restarting the work cycle. In event time, these RDOs fall on days -5, 0, and 1. In Figure 2a, we plot the proportion of treatment and control officers reporting for work in each day of our sub-experiments. Treatment and control officers are equally likely to report for work on scheduled work days ($t = -4, \dots, -1$ and $t = 2, \dots, 5$), and are both roughly 50 percentage points less likely to report for work on a scheduled and granted day off ($t = 1$). However, while control group attendance rates drop similarly on the canceled RDO ($t = 0$), about half of the treatment group reports to work. This corresponds to treatment officers attending work at a rate that is approximately 25 percentage points higher than control officers on the date of treatment. This estimate is virtually unchanged when we formalize the comparison in our event study specification. In Figure 2b, we estimate that being assigned treatment yields a statistically significant increase of about 25 percentage points in terms of an officer’s probability of coming into work on the day of the cancellation.

¹⁰Importantly, as we describe in section 3 and show in section C.1, assignment switching is not observed in anticipation of a canceled RDO.

The ratio of officers calling in sick to those present at work remains stable between pre-treatment periods and the canceled day off, as shown in Figure 3c. This stability affirms that there are no anticipatory effects from the policy and that illness is not a significant factor in policy compliance. With respect to the intensive margin, Figure 2c and Figure 2d display binscatter and event-study plots of the number of overtime hours an officer works on a given day. Both plots show that on canceled RDOs, treated officers work an average of approximately 2.5 additional hours compared to the control group. This increase is roughly equal to a standard 9-hour shift multiplied by the compliance rate, which implies that officers who comply with the policy work a full shift on their canceled RDO. In total, we find strong evidence that a canceled RDO induces the anticipated effect of working on a RDO.

4.2 Attendance and Officer Wellness

A primary criticism of mandatory overtime, and a key criticism that motivated CPD to discontinue the policy, is the concern that it undermines officer wellness. One might expect this concern to be particularly salient for physically demanding jobs like that of a police officer and assignments within policing like that of patrol. To evaluate this possibility, we use detailed attendance records to examine whether canceled RDOs affect direct measures of wellness—such as illness or injury—as well as compensatory behaviors like reducing subsequent time at work. Panel A of Table 2 reports these results.

The raw data provide little evidence of compensatory behavior. On the pre-treatment workdays ($t = -4, \dots, -1$), Figure 2a displays nearly identical attendance rates for treatment and control officers, which diverge sharply on the canceled RDO due to compliance. However, we do not observe any meaningful difference in the post-treatment period. On subsequent workdays after the canceled RDO, treated officers are no more likely than controls to take additional days off. Similarly, Figure 2c shows no decline in overtime hours in the days following a canceled RDO. Turning to absences, Figure 3a indicates that injury-related absences are highly stable across groups and time, while sick day rates in Figure 3c fluctuate more but maintain parallel differences between treatment and control officers into the post-treatment periods.

Event study estimates confirm these patterns. As shown in Figure 2b and Figure 2d, treatment and control groups exhibit parallel pre-trends and no evidence of compensatory behavior after the canceled RDO. Likewise, Figure 3b and Figure 3d show that injury and sick absences remain flat and do not deviate meaningfully from pre-treatment trajectories in the post-period.

The aggregated difference-in-differences estimates in Panel A of Table 2 provide precise evidence of null effects. Officers do not compensate for canceled RDOs by missing subsequent workdays (-0.35 pp, insignificant) or by reducing their overtime hours (a negligible increase of 2.4 minutes per day). Nor do we observe any significant increase in absences due to injury or illness: injury absences rise by only 0.04 pp (about 1 percent of the pre-period mean), while sick absences actually decline by 0.18 pp (insignificant). The confidence intervals for these estimates rule out economically meaningful increases, suggesting that mandatory overtime does not compromise officer attendance or health.

4.3 Officer Productivity

Although officers continue to show up for work following a canceled RDO, mandatory overtime may still affect how productive they are when they are on patrol. To test this possibility, we examine two central measures of patrol productivity: calls for service (CFS), which capture an officer’s response to demand for service, and investigatory stops, which capture discretionary officer activity.

The binscatter plots in Figure 4a and Figure 4c show that treatment and control officers had nearly identical pre-treatment patterns in both outcomes. The one exception is a slight divergence in investigatory stops just prior to treatment, but this difference is economically small, amounting to fewer than 0.01 stops per officer per day.

Dynamic event study estimates, presented in Figure 4b and Figure 4d, confirm that pre-treatment trends are parallel and that treatment has no post-period effects. The daily treatment effects are small and stable across the entire work cycle following the canceled RDO.

Aggregated difference-in-differences estimates, reported in Panel A of Table 2, columns 5 and 6, corroborate these findings. Assignment to a canceled RDO has no statistically or economically significant effect on productivity: officers respond to 0.003 additional calls per day (95% CI: $-0.061, 0.068$) and initiate 0.002 fewer stops per day (2.3 percent of the mean, with the 95% CI ruling out decreases larger than 0.007). These results provide strong evidence that mandatory overtime does not affect the quantity of officers’ on-the-job activity.

4.4 Enforcement Activity

Finally, we examine whether canceled RDOs affect officer enforcement behavior, specifically arrests and the use of force. These outcomes represent a more consequential margin of officer behavior and are directly linked to public safety.

The raw data suggest little evidence of systematic differences between treatment and control officers. As shown in Figure 5a and Figure 5c, about 10 percent of officers make at least one arrest on a given workday, while about 0.2 percent use force. These rates and trends are similar across treatment and control groups prior to a cancellation, with the exception of an idiosyncratic fluctuation in use of force at $t = -3$. As noted in Section 3, this appears to be noise in a single period rather than evidence of a violated parallel trends assumption.

The dynamic event study plots in Figure 5b and Figure 5d show no consistent differences between treatment and control groups in either arrests or use of force. Pre-treatment trends track closely, and post-treatment estimates remain flat, indicating that mandatory overtime does not affect the likelihood of enforcement actions.

Aggregated difference-in-differences estimates, presented in Panel A of Table 2, columns 7 and 8, confirm these results. Officers assigned to a canceled RDO are 0.16 pp more likely to make an arrest in the post-period, an effect that is statistically insignificant and corresponds to just 1.7 percent of the pre-treatment mean. The probability of using force falls by 0.01 pp, also statistically insignificant. Disaggregated results by arrest type, reported in Table 3, show no evidence of substitution across offense categories. Insignificant changes range from 1.6 percent to -3.9 percent of baseline means for index, traffic, and other arrests. The only large proportional effect arises for arrests without charges, which are rare (0.4 percent of officer-days). Here, the point estimate suggests a 0.04 pp increase, or 10.5 percent of the mean, but given the low base rate, we regard this as economically negligible. Overall, the evidence indicates that mandatory overtime does not alter officers' enforcement practices.

4.5 Robustness

Our main findings are robust to alternative specifications and sampling choices. We begin by varying the fixed effects structure. Because RDO cancellations are imposed at the DOG-area-date level, our baseline specification includes officer, DOG-by-date, and area-by-date fixed effects. As shown in Table B.4, the results are unchanged when simplifying the model by dropping officer fixed effects, and

they remain stable under a more conservative specification that additionally includes DOG-by-area fixed effects.

We next consider the possibility that treatment effects persist for longer than a single work cycle. If so, our control group could be contaminated by recently treated officers, attenuating our estimates. To address this concern, we re-estimate our models using a restricted sample that excludes officers treated within the prior two work cycles. The results, reported in Table B.6 and Table B.7, are substantively similar to our baseline specification and generally characterized by small, insignificant effects across domains. The only exceptions are modest increases in investigatory stops and “other” arrests, though even these are small in absolute terms. The corresponding event study plots in Appendix section C.3 confirm that these findings are consistent over time.

Finally, we examine whether treatment effects might emerge with a delay, in which case our short post-treatment window could understate the true policy impact. To test this, we extend the event window to include two full work cycles before and after each cancellation, restricting to officers not treated within that horizon. As reported in Table B.8 and Table B.9, the difference-in-differences estimates for attendance, wellness, productivity, and enforcement remain uniformly small and statistically insignificant. Appendix section C.4 presents the associated binscatter and event study plots, which corroborate the null results and show that temporal aggregation does not mask meaningful dynamics across periods, aside from a modest decline in enforcement in the second work cycle.

Together, these robustness checks confirm that our findings are not sensitive to fixed effects structure, assumptions about the persistence of treatment effects, or the length of the post-treatment window.

4.6 Treatment Heterogeneity

In the preceding sections, we described empirical evidence that the policy did not negatively impact officers across a range of measures. However, it is possible that officers with different characteristics experience the policy differently and that these heterogeneous effects are obscured when averaging over all officers. This possibility is both theoretically and empirically grounded. A large literature documents how worker productivity varies with experience, broadly, due both to selection and learning-by doing (Haggag et al. (2017), Daveri and Parisi (2015), Papay and Kraft (2015)), and police productivity specifically differs by demographic and tenure characteristics (Ba et al. (2021), Ba et al. (2022)). Under mandatory overtime, workers more susceptible to burnout have limited ability to opt out of

the additional hours. For this reason, it is useful to characterize officer traits that predict larger effects.

We test for heterogeneity in the effects of mandatory overtime over a range of officer characteristics including demographics, experience, productivity, and history of canceled RDOs. We focus on these characteristics because we hypothesize that they could affect treatment compliance and susceptibility to burnout. For example, more senior officers may be better able to evade treatment, attenuating effects more than for junior officers. Officers who respond to fewer calls for service, a measure of their productivity, may be more likely to hold leadership roles and less likely to be on patrol. Then RDO cancellations may not apply to them, or if they do, they may experience less strain from working in the field in the prior week. Moreover, burnout is likely to be the result of cumulative job-related stress. RDO cancellations that occur at the start of our study period may be too early to show evidence of burnout, as officers would have experienced treatment fewer times. Alternatively, if it takes time for officers to adjust to the policy change, it is possible we may see more evidence of a negative effect on officer behavior early on compared to when they have experienced many RDO cancellations.

In our heterogeneity analysis, we split the sample by officer characteristics and estimate the prior difference-in-differences specification using each of the resulting 11 sub-samples. We present the estimated treatment effects for each subgroup in Figure 6 and Figure 7.¹¹ For example, we show the effect of being assigned RDO cancellation on officer attendance for the subsample of male, and the subset of female, officers separately.

In Figure 6, we plot the coefficients and 95% confidence intervals from the split samples for each of our attendance and wellness outcomes: attendance, overtime hours, injuries, and sick days. Broadly, results across subgroups and outcomes support the null results from our primary analyses. No group exhibited statistically significant or economically large changes in the probabilities they reported to work, called out injured, or called out sick. Echoing our primary results, we find small, positive point estimates for effects on overtime hours for all groups – most, but not all of which are statistically significant. While we observe some variability in the point estimates across groups, we nonetheless observe considerable parity in point estimates for natural comparison groups: when comparing effects for male to female officers, White to Hispanic and Black officers, etc. we find qualitatively similar results. The rare exception to this

¹¹Officer demographics are directly reported in rosters. Due to small sample sizes, we do not include results for officers who identify as Asian, Native/Pacific Islander, or Other. We define the categories of officer experience, tenure, and treatment history based on the median officer in our sample.

pattern is in overtime hours, where officers treated fewer times exhibit a noticeably larger increase in overtime hours than their more-often treated counterparts.

Figure 7 presents subgroup treatment effects for enforcement and productivity-related outcomes. Across these domains, our estimates are somewhat noisier than the attendance and wellness results, but show similar trends: we observe small, statistically insignificant point estimates for nearly all outcomes and subgroups, with results congruent across subgroups. Again, we note one exception to this pattern, this time in the “Any Arrests” outcome. Whereas Hispanic officers show a moderate increase in arrest propensities following treatment, Black officers exhibit a moderate decrease. Because this finding is not consistent with a broader pattern of responses for Hispanic or Black officers, we are inclined to attribute it to happenstance. Officers of various demographic backgrounds, experience levels, productivity patterns, and treatment histories demonstrate stable wellness, workplace engagement, productivity, and enforcement outcomes after mandatory overtime.

5 Conclusion

In this paper, we evaluate the effects of a mandatory overtime policy on patrol officer health and performance, studying a unique policy implemented by the Chicago Police Department (CPD) from 2022 to 2023. This policy addressed chronic staffing shortages by systematically canceling officers’ scheduled days off in a six-week rotation, thereby increasing their work hours. Using a stacked difference-in-differences design, which leverages plausibly exogenous variation in these cancellations, we find robust evidence that the policy had no significant effect on a broad range of outcomes.

Our analysis shows that a single mandatory overtime shift does not meaningfully affect officer wellness, productivity, or enforcement activities. We find no evidence that the policy led to negative health effects, such as an increase in injury or illness-related absences, and our estimates are precise enough to rule out even modest changes in most outcomes. Similarly, we find no detrimental effects on the public, as officers’ productivity—measured by their response to calls for service and investigatory stops—remained unchanged.

We offer two potential explanations for these precise null effects, which likely operate in tandem. First, the magnitude of the intervention was modest. The policy imposed a single mandatory workday, resulting in a five-days-on, one-day-off cycle for affected officers. At an individual level, officers were assigned to mandatory overtime no more than once a month. While it is reasonable to

anticipate that sufficiently large or sustained workload increases could alter officer behavior and wellness, our findings do not preclude the existence of such a “tipping point” for fatigue or burnout. Instead, they suggest that temporary increases in workload, particularly those followed by a rest day, may not be sufficient to induce a measurable decline in performance. We study only the short-run effect of this policy and note that it is possible that there are deleterious effects that build over the long-run.

Second, the characteristics of the work performed on these overtime days may have contributed to our results. We observe that officers respond to fewer calls for service and make fewer stops and arrests on a canceled RDO than on a typical workday. Thus, it appears that supervisors preemptively assign or that officers self-select into lower-intensity duties (such as the visible deterrence duties of “Operation Scarecrow”) when working a canceled RDO. This form of substitution could be a key mechanism through which the policy avoids potentially detrimental effects on officers and the public.

In summation, our findings suggest that a limited mandatory overtime policy can be a useful short-run tool for police departments and other organizations facing significant labor constraints. Because we find precise null effects, our research also indirectly suggests that equally modest reductions in workload are unlikely to produce substantial improvements in frontline worker performance or well being. For public service provision, decisions regarding changes to marginal workloads may be better evaluated on the basis of aggregate changes and organizational needs rather than the expectation of meaningful changes to employee wellness or performance.

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6 Tables

Table 1: Summary Statistics

VARIABLES	Linked Sample		Stacked Sample	
	Mean	SD	Mean	SD
Reports to Work	.5382	.4985	.5455	.4979
Total Overtime Hours	.9277	2.551	.7835	2.305
Total Involuntary Overtime Hours	.1952	1.298	.1636	1.198
Total Calls for Service	1.88	3.512	2.373	3.856
Total Investigatory Stops	.04466	.316	.05971	.3663
Any Arrests	.0587	.2351	.07468	.2629
Any Use of Force	.0012	.0356	.001749	.04287
Female	.2037	.4028	.2202	.4144
White	.6845	.4647	.6838	.465
Hispanic	.346	.4757	.3872	.4871
Black	.1477	.3548	.1649	.3711
Years of Experience	12.29	8.869	9.843	8.053
N	3,152,660		1,854,336	
Injured on the Job	.03508	.184	.04412	.2054
Calls in Sick	.05634	.2306	.06146	.2402
N	3,147,027		1,850,441	
Distinct Officers	7,927	-	5,864	-
Distinct Sub-Experiments	-	-	288	-
Officers per Sub-Experiment	-	-	623.4	146.8

Table 2: Main Outcomes

	(A) Attendance and Work		(B) Officer Wellness	
Dependent Variable:	(1) Reports to Work	(2) Total Overtime Hours	(3) Injured on the Job	(4) Calls in Sick
Treatment $\times$ Post	-0.0035 (0.0040)	0.0433 (0.0142)	0.0004 (0.0009)	-0.0018 (0.0019)
Pre-Period Mean	0.6735	0.5040	0.0444	0.0671
Observations	1348608	1348608	1345779	1345779
	(C) Officer Productivity		(D) Enforcement Activity	
Dependent Variable:	(5) Total Calls for Service	(6) Total Investigatory Stops	(7) Any Arrests	(8) Any Use of Force
Treatment $\times$ Post	0.0034 (0.0328)	-0.0018 (0.0027)	0.0016 (0.0023)	-0.0001 (0.0002)
Pre-Period Mean	3.0738	0.0779	0.0965	0.0021
Observations	1348608	1348608	1348608	1348608

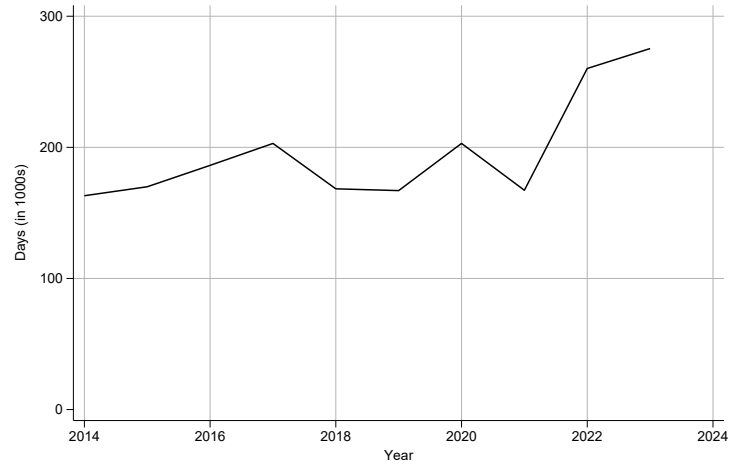
Data are restricted to exclude observations falling on RDOs, specifically with event-times $t \in \{-5, 0, 1\}$. Results are estimated using an aggregated version of our dynamic event-study specification, Equation 1. We report the coefficients on interaction between treatment and a post-treatment indicator, omitting coefficients for event-times $t < 1$. Standard errors clustered by area-DOG are reported in parentheses. The pre-treatment mean of each outcome is calculated for event-times $t < 0$, excluding RDOs.

Table 3: Arrest Type

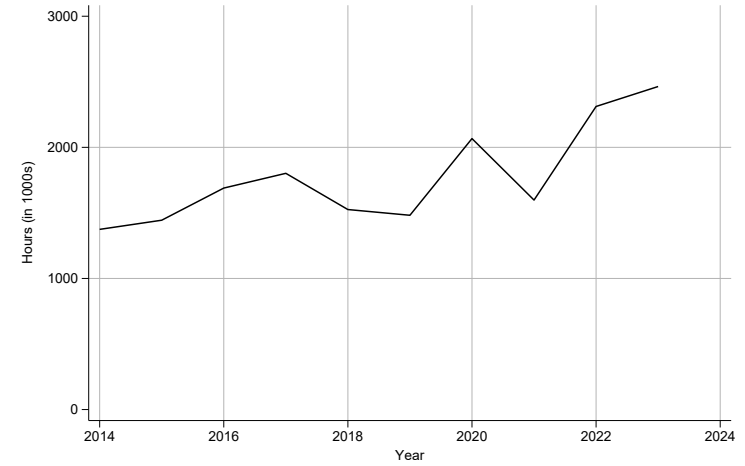
Dependent Variable:	(1) Any Index Arrests	(2) Any Traffic Arrests	(3) Other Arrests	(4) Any Arrests with No Charge
Treatment $\times$ Post	-0.0006 (0.0011)	-0.0011 (0.0015)	0.0013 (0.0018)	0.0004 (0.0006)
Pre-Period Mean	0.0196	0.0283	0.0837	0.0038
Observations	1348608	1348608	1348608	1348608

Data are restricted to exclude observations falling on RDOs, specifically with event-times $t \in \{-5, 0, 1\}$. Results are estimated using an aggregated version of our dynamic event-study specification, Equation 1. We report the coefficients on interaction between treatment and a post-treatment indicator, omitting coefficients for event-times $t < 1$. Standard errors clustered by area-DOG are reported in parentheses. The pre-treatment mean of each outcome is calculated for event-times $t < 0$, excluding RDOs.

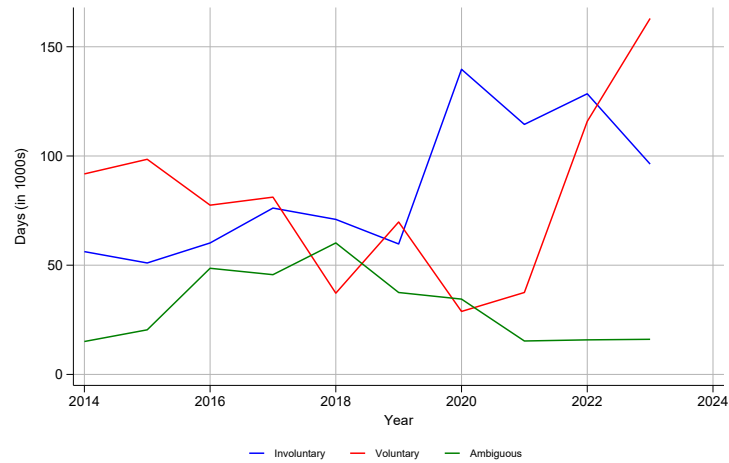
7 Figures



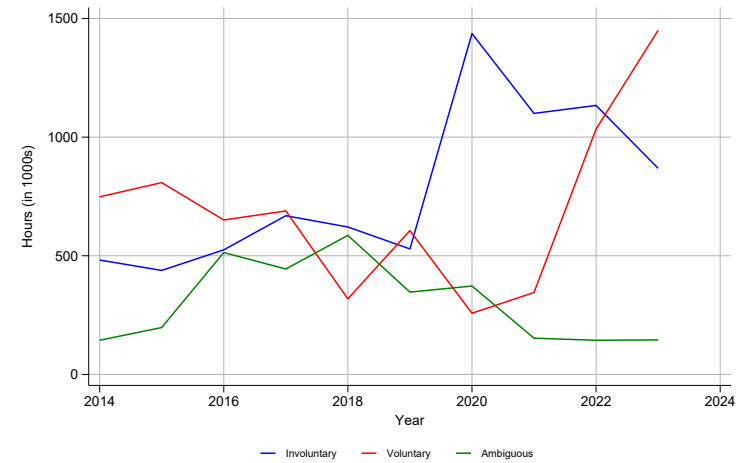
(a) Cumulative Days with Overtime



(b) Cumulative Hours of Overtime



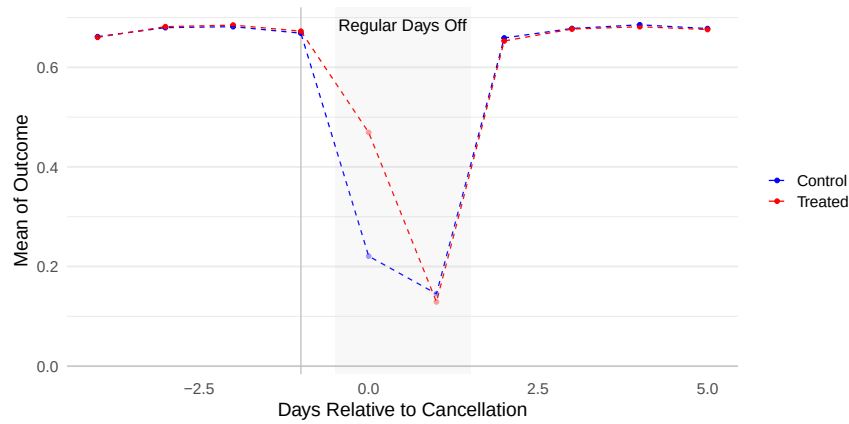
(c) Cumulative Days with Overtime by Type



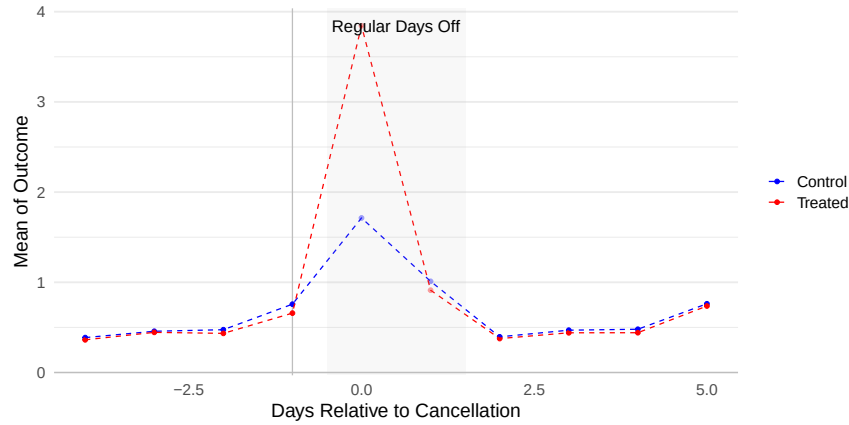
(d) Cumulative Hours of Overtime by Type

Figure 1: Trends in Overtime from 2014 to 2023

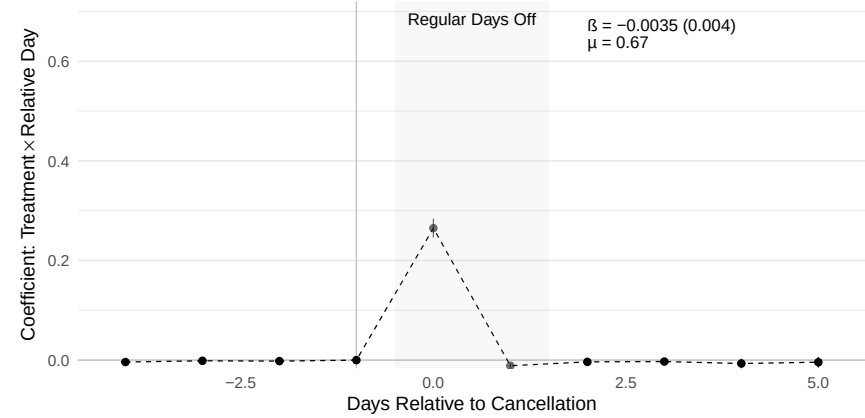
These figures show trends in overtime usage per officer. The data is restricted to overtime assignments where an officer worked 8 or more hours. Panel (a) plots cumulative days with an officer taking any overtime since 2014, panel (b) plots cumulative hours of overtime since 2014, panel (c) plots cumulative days of overtime by type since 2014, and panel (d) plots cumulative hours of overtime by type since 2014.



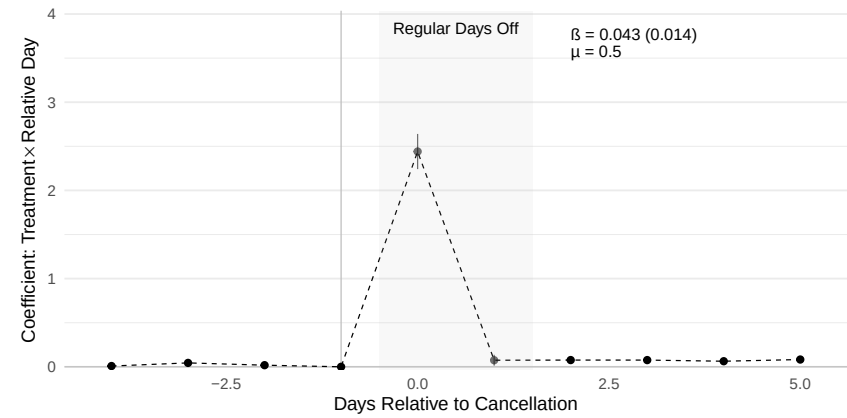
(a) Raw Data: Reported to Work



(c) Raw Data: Overtime Hours



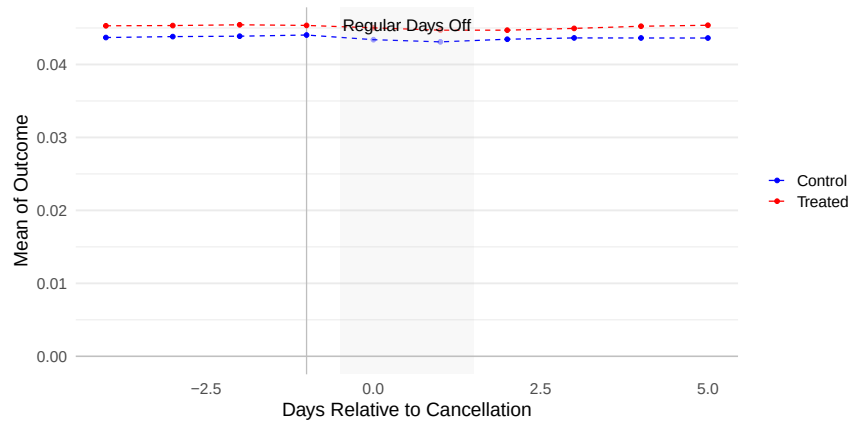
(b) Event Study: Reported to Work



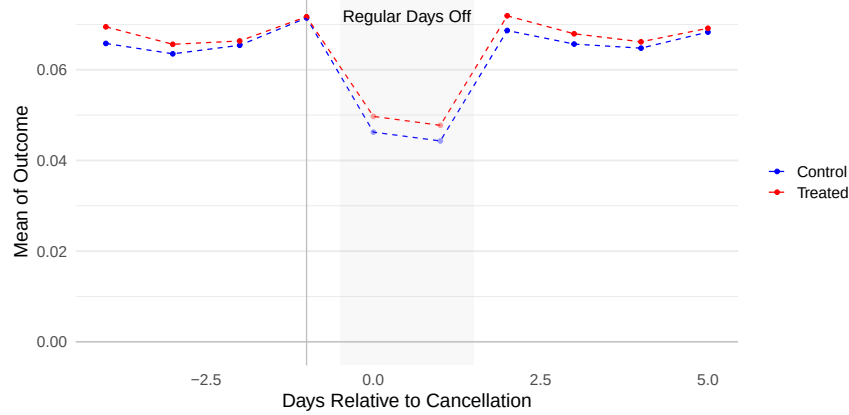
(d) Event Study: Overtime Hours

Figure 2: Treatment Compliance and Effects on Attendance

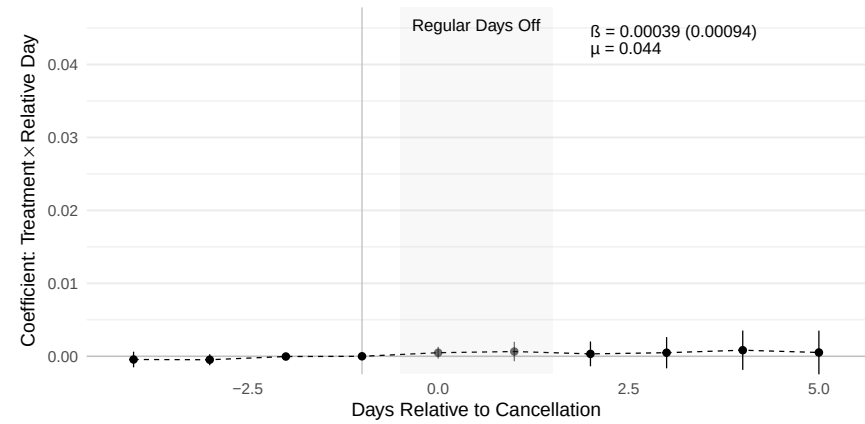
These figures show the effect of overtime assignments on daily officer attendance (panels a, b) and overtime hours (c, d). Panels (a) and (c) present raw binscatter evidence, while panels (b) and (d) display event study estimates from Equation 1. Event study specifications include officer, DOG-date, and Area-date fixed effects, with coefficients interpreted relative to the day before treatment. Standard errors clustered by Area-DOG.



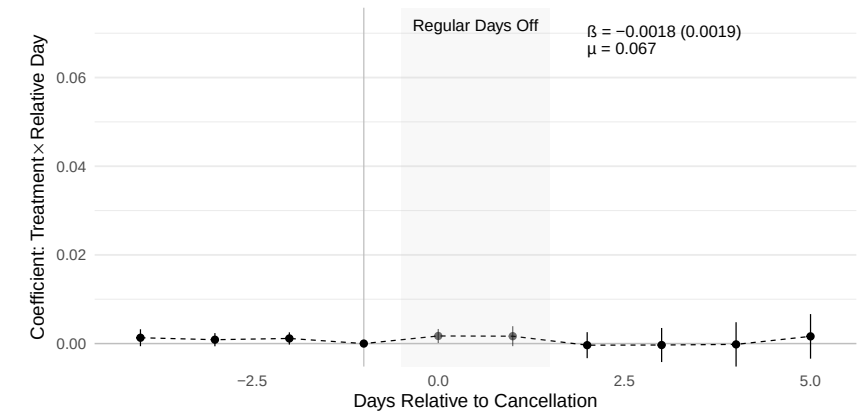
(a) Raw Data: Injury



(c) Raw Data: Calls in Sick



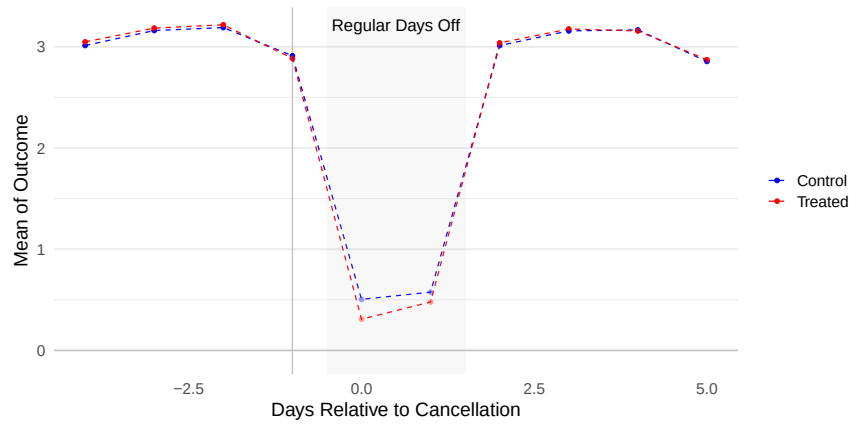
(b) Event Study: Injury



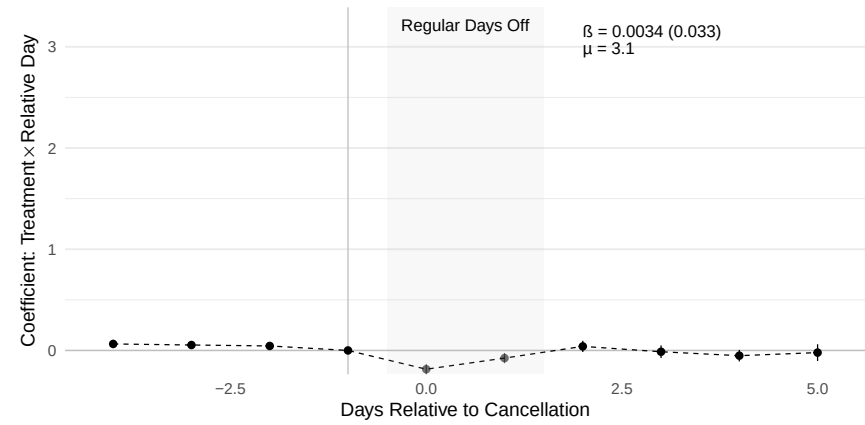
(d) Event Study: Calls in Sick

Figure 3: Effects on Officer Wellness

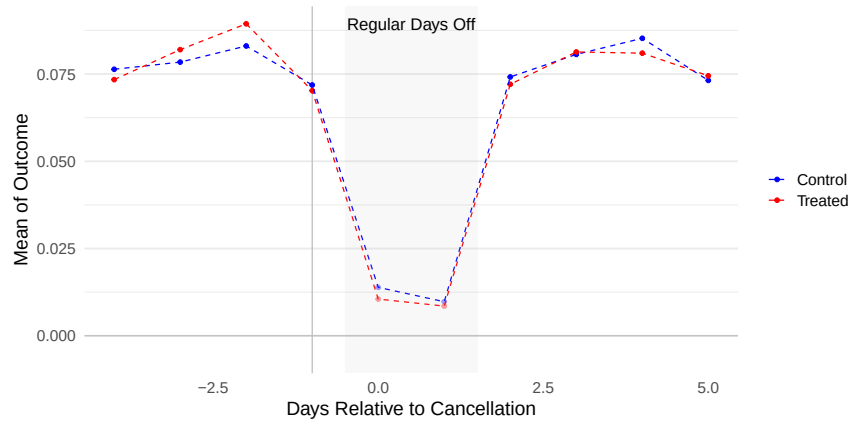
These figures show the effect of overtime assignments on officer wellness. Panels (a) and (c) present raw binscatter evidence, while panels (b) and (d) display event study estimates from Equation 1. Event study specifications include officer, DOG-date, and Area-date fixed effects, with coefficients interpreted relative to the day before treatment. Standard errors clustered by Area-DOG.



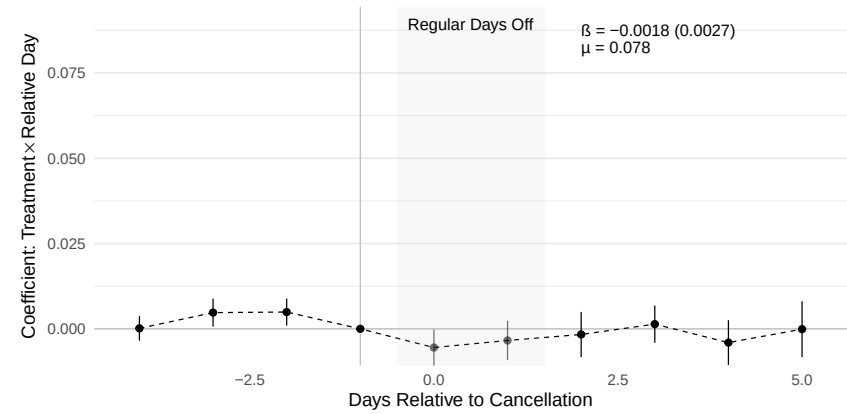
(a) Raw Data: Calls for Service



(b) Event Study: Calls for Service



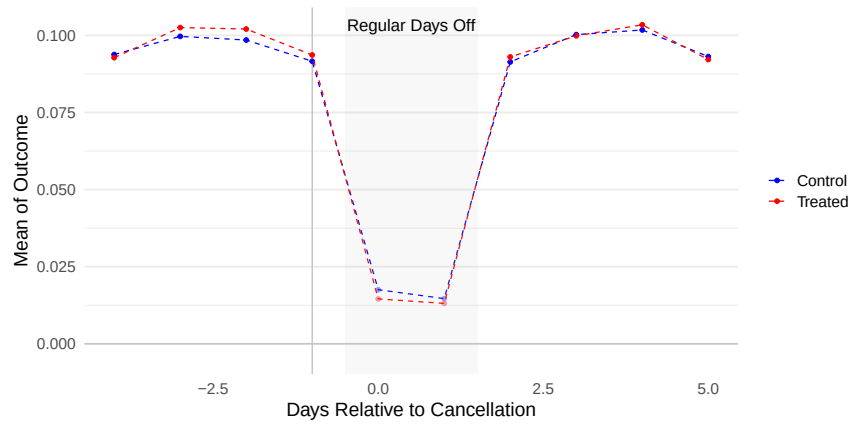
(c) Raw Data: Stops



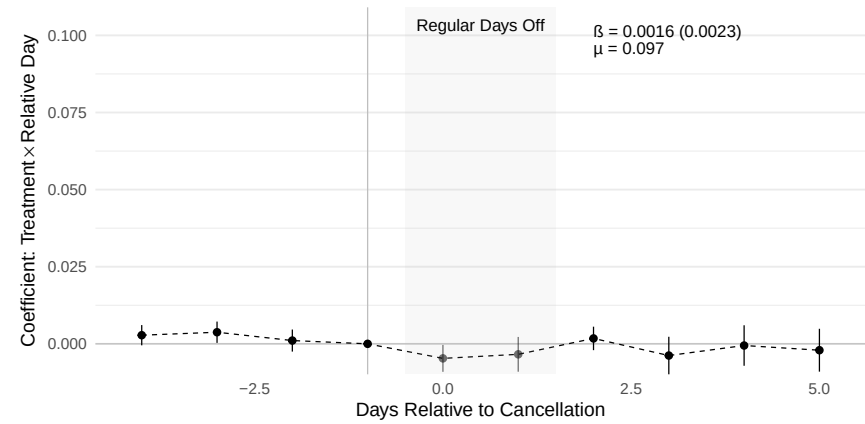
(d) Event Study: Stops

Figure 4: Effects on Productivity

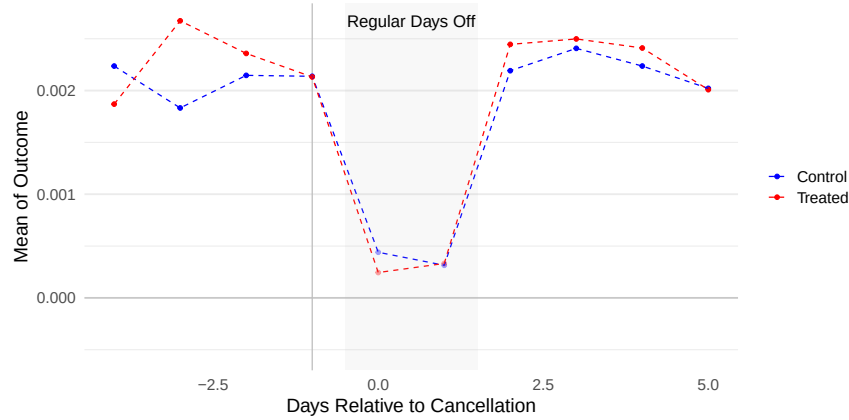
These figures show the effect of overtime assignments on officer productivity. Panels (a) and (c) present raw binscatter evidence, while panels (b) and (d) display event study estimates from Equation 1. Event study specifications include officer, DOG-date, and Area-date fixed effects, with coefficients interpreted relative to the day before treatment. Standard errors clustered by Area-DOG.



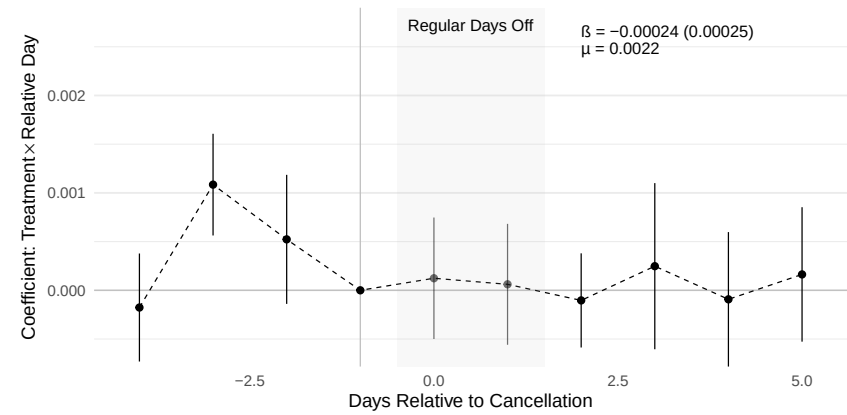
(a) Raw Data: Arrests



(b) Event Study: Arrests



(c) Raw Data: Use of Force



(d) Event Study: Use of Force

Figure 5: Effects on Law Enforcement Activity

These figures show the effect of overtime assignments on officer enforcement activities. Panels (a) and (c) present raw binscatter evidence, while panels (b) and (d) display event study estimates from Equation 1. Event study specifications include officer, DOG-date, and Area-date fixed effects, with coefficients interpreted relative to the day before treatment. Standard errors clustered by Area-DOG.

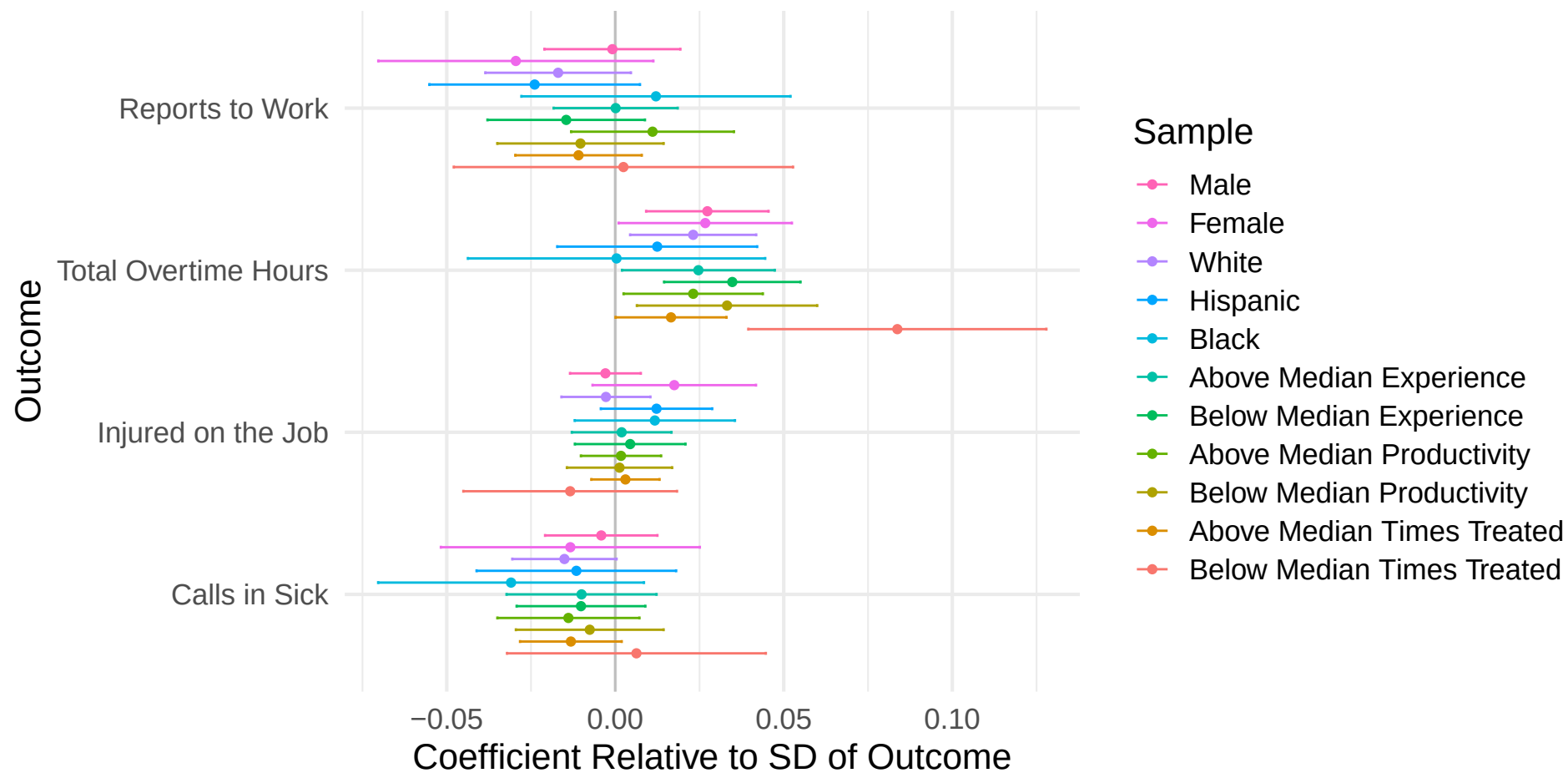


Figure 6: Attendance and Wellness

These figures show how the effects of overtime assignments vary across officer subgroups. Each coefficient plot is based on separate difference-in-differences regressions modified from Equation 1 estimated within subgroup categories, illustrating heterogeneous effects across attendance and wellness outcomes.

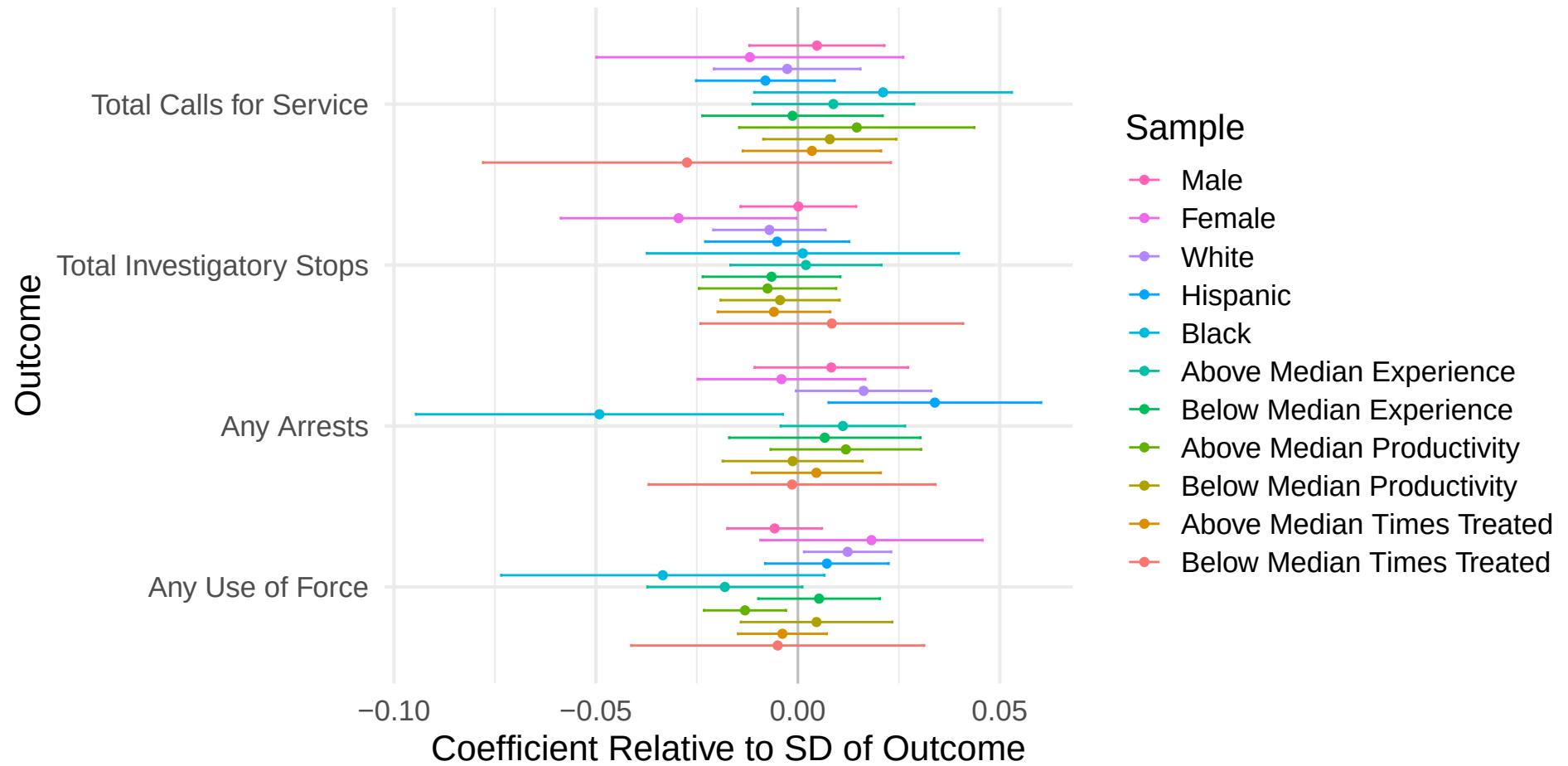


Figure 7: Enforcement and Productivity

These figures show how the effects of overtime assignments vary across officer subgroups. Each coefficient plot is based on separate difference-in-differences regressions modified from Equation 1 estimated within subgroup categories, illustrating heterogeneous effects across productivity and enforcement outcomes.

A Data

We analyze data from CPD officers in DOGs 61-66, which includes approximately 66% of officer-date combinations we observe in a linked A&A and officer roster data set. We restrict our sample to this subset of DOGs because they adhered to the stable six day work cycle described in the main text, and were subject to the RDO cancellation policy of interest. We also restrict our sample to officers assigned areas numbered 1-5. While The CPD does assign some officers to alternative “areas”, these assignments are mainly administrative, are not associated with a specific geographic area, and so the job tasks fall outside our productivity measures of interest.

We identify RDO cancellations using cancellation matrices obtained from the CPD. Figure A.8 shows an example of an RDO cancellation matrix we obtained from the CPD through a FOIA request. RDO cancellations have a sequential, cyclic structure, with occasional exceptions. For example, on May 31, 2022, the cancellation cycle begins with officers in Area 1 and DOG 61¹². Comparing this figure to the operations calendar for 2022 confirms that May 31st is the first of two regularly scheduled days-off for DOG 61. The following day, June 1st, 2022, officers in Area 1 in DOG 62 were assigned an RDO cancellation and so on until by June 5th, 2022, all officers in DOGs 61 to 66 in Area 1 had been assigned treatment. Next, on June 7th, officers in all DOGs 61 to 66 in Areas 3 and 4 were sequentially assigned treatment, starting with DOG 62.

¹²DOG 75 also experiences an RDO cancellation on this date, but our analysis is restricted to officers in DOGs 61 to 66. See 2.2 for details on this sample restriction.

Dates	RDO	District Law Enforcement				
		Area 1	Area 2	Area 3	Area 4	Area 5
Tues, 31-May	61, 75	X				
Wed, 01-Jun	62, 76	X				
Thurs, 02-Jun	63, 77	X				
Fri, 3-Jun	64, 71, F/S	X				
Sat, 4-Jun	65, 72, S/S	X				
Sun, 5-Jun	66, 73, S/M	X				
Tues, 07-Jun	62, 74			X	X	
Wed, 08-Jun	63, 75			X	X	
Thurs, 09-Jun	64, 76			X	X	
Fri, 10-Jun	65, 77, F/S			X	X	
Sat, 11-Jun	66, 71, S/S			X	X	
Sun, 12-Jun	61, 72, S/M			X	X	
Tues, 21-Jun	64, 72		X			X
Wed, 22-Jun	65, 73		X			X
Thurs, 23-Jun	66, 74		X			X
Fri, 24-Jun	61, 75, F/S		X			X
Sat, 25-Jun	62, 76, S/S		X			X
Sun, 26-Jun	63, 77, S/M		X			X
Tues, 05-Jul	66, 77	X				
Wed, 06-Jul	61, 71	X				
Thurs, 07-Jul	62, 72	X				
Fri, 8-Jul	63, 73, F/S	X				
Sat, 9-Jul	64, 74, S/S	X				
Sun, 10-Jul	65, 75, S/M	X				
Tues, 12-Jul	61, 76			X	X	
Wed, 13-Jul	62, 77			X	X	
Thurs, 14-Jul	63, 71			X	X	
Fri, 15-Jul	64, 72, F/S			X	X	
Sat, 16-Jul	65, 73, S/S			X	X	
Sun, 17-Jul	66, 74, S/M			X	X	
Tues, 19-Jul	62, 75		X			X
Wed, 20-Jul	63, 76		X			X
Thurs, 21-Jul	64, 77		X			X
Fri, 22-Jul	65, 71, F/S		X			X
Sat, 23-Jul	66, 72, S/S		X			X
Sun, 24-Jul	61, 73, S/M		X			X

Figure A.8: Example CPD Cancellation Matrix

B Tables

Table B.4: Main Sample: No Officer Fixed-Effects

	<i>(A) Attendance and Work</i>		<i>(B) Officer Wellness</i>	
Dependent Variable:	(1) Reports to Work	(2) Total Overtime Hours	(3) Injured on the Job	(4) Calls in Sick
Treatment $\times$ Post	-0.0015 (0.0040)	0.0412 (0.0142)	-0.0006 (0.0014)	-0.0026 (0.0024)
Pre-Period Mean	0.6735	0.5040	0.0444	0.0671
Observations	1348608	1348608	1345779	1345779
	<i>(C) Officer Productivity</i>		<i>(D) Enforcement Activity</i>	
Dependent Variable:	(5) Total Calls for Service	(6) Total Investigatory Stops	(7) Any Arrests	(8) Any Use of Force
Treatment $\times$ Post	0.0121 (0.0291)	-0.0036 (0.0032)	0.0016 (0.0023)	-0.0001 (0.0002)
Pre-Period Mean	3.0738	0.0779	0.0965	0.0021
Observations	1348608	1348608	1348608	1348608

Data are restricted to exclude observations falling on RDOs, specifically with event-times $t \in \{-5, 0, 1\}$. Results are estimated using an aggregated version of our dynamic event-study specification, Equation 1. Unlike Equation 1, we do not include officer fixed effects in this specification. We report the coefficients on interaction between treatment and a post-treatment indicator, omitting coefficients for event-times $t < 1$.

Standard errors clustered by area-DOG are reported in parentheses. The pre-treatment mean of each outcome is calculated for event-times $t < 0$, excluding RDOs.

Table B.5: Main Sample: Including Area-DOG Fixed-Effects

	<i>(A) Attendance and Work</i>		<i>(B) Officer Wellness</i>	
Dependent Variable:	(1) Reports to Work	(2) Total Overtime Hours	(3) Injured on the Job	(4) Calls in Sick
Treatment $\times$ Post	-0.0034 (0.0039)	0.0433 (0.0141)	0.0002 (0.0010)	-0.0018 (0.0018)
Pre-Period Mean	0.6735	0.5040	0.0444	0.0671
Observations	1348608	1348608	1345779	1345779
	<i>(C) Officer Productivity</i>		<i>(D) Enforcement Activity</i>	
Dependent Variable:	(5) Total Calls for Service	(6) Total Investigatory Stops	(7) Any Arrests	(8) Any Use of Force
Treatment $\times$ Post	0.0006 (0.0331)	-0.0015 (0.0026)	0.0014 (0.0023)	-0.0001 (0.0002)
Pre-Period Mean	3.0738	0.0779	0.0965	0.0021
Observations	1348608	1348608	1348608	1348608

We exclude observations falling on RDOs, specifically with event-times $t \in \{-5, 0, 1\}$. Results are estimated using an aggregated version of our dynamic event-study specification, Equation 1. Unlike Equation 1, we additionally include Area-DOG fixed effects in this specification. We report the coefficients on interaction between treatment and a post-treatment indicator, omitting coefficients for event-times $t < 1$. Standard errors clustered by area-DOG are reported in parentheses. The pre-treatment mean of each outcome is calculated for event-times $t < 0$, excluding RDOs.

Table B.6: Main Results: Restricted Sample

	<i>(A) Attendance and Work</i>		<i>(B) Officer Wellness</i>	
Dependent Variable:	(1) Reports to Work	(2) Total Overtime Hours	(3) Injured on the Job	(4) Calls in Sick
Treatment $\times$ Post	-0.0070 (0.0043)	0.0503 (0.0150)	-0.0004 (0.0021)	0.0022 (0.0027)
Pre-Period Mean	0.6727	0.5075	0.0443	0.0663
Observations	1028264	1028264	1026227	1026227
	<i>(C) Officer Productivity</i>		<i>(D) Enforcement Activity</i>	
Dependent Variable:	(5) Total Calls for Service	(6) Total Investigatory Stops	(7) Any Arrests	(8) Any Use of Force
Treatment $\times$ Post	0.0601 (0.0364)	-0.0126 (0.0048)	0.0040 (0.0034)	-0.0002 (0.0003)
Pre-Period Mean	3.0658	0.0785	0.0966	0.0022
Observations	1028264	1028264	1028264	1028264

In this specification, we exclude observations falling on RDOs. We additionally omit all officers who were treated within two work cycles prior to the focal treatment. Results are estimated using an aggregated version of our dynamic event-study specification, Equation 1. We report the coefficients on interaction between treatment and a post-treatment indicator, omitting coefficients for event-times $t < 1$. Standard errors clustered by area-DOG are reported in parentheses. The pre-treatment mean of each outcome is calculated for event-times $t < 0$, excluding RDOs.

Table B.7: Arrest Type: Restricted Sample

Dependent Variable:	(1) Any Index Arrests	(2) Any Traffic Arrests	(3) Other Arrests	(4) Any Arrests with No Charge
Treatment $\times$ Post	-0.0009 (0.0013)	0.0006 (0.0016)	0.0062 (0.0025)	0.0002 (0.0006)
Pre-Period Mean	0.0195	0.0285	0.0836	0.0037
Observations	1028264	1028264	1028264	1028264

In this specification, we exclude observations falling on RDOs and omit all officers who were treated within two work cycles prior to the focal treatment. Results are estimated using an aggregated version of our dynamic event-study specification, Equation 1. We report the coefficients on interaction between treatment and a post-treatment indicator, omitting coefficients for event-times $t < 1$. Standard errors clustered by area-DOG are reported in parentheses. The pre-treatment mean of each outcome is calculated for event-times $t < 0$, excluding RDOs.

Table B.8: Main Results: Long Sample

	<i>(A) Attendance and Work</i>		<i>(B) Officer Wellness</i>	
Dependent Variable:	(1) Reports to Work	(2) Total Overtime Hours	(3) Injured on the Job	(4) Calls in Sick
Treatment $\times$ Post	-0.0107 (0.0063)	0.0380 (0.0196)	-0.0016 (0.0017)	-0.0026 (0.0020)
Pre-Period Mean	0.6733	0.5575	0.0408	0.0642
Observations	1061472	1061472	1060274	1060274
	<i>(C) Officer Productivity</i>		<i>(D) Enforcement Activity</i>	
Dependent Variable:	(5) Total Calls for Service	(6) Total Investigatory Stops	(7) Any Arrests	(8) Any Use of Force
Treatment $\times$ Post	0.0149 (0.0488)	0.0019 (0.0049)	0.0048 (0.0034)	-0.0005 (0.0003)
Pre-Period Mean	3.0096	0.0780	0.0959	0.0021
Observations	1061472	1061472	1061472	1061472

In this specification, we exclude observations falling on RDOs. We additionally include observations from the two work cycles before and after the focal treatment – and omit all control group officers who were treated within these four work cycles. Results are estimated using an aggregated version of our dynamic event-study specification, Equation 1. We report the coefficients on interaction between treatment and a post-treatment indicator, omitting coefficients for event-times $t < 1$. Standard errors clustered by area-DOG are reported in parentheses. The pre-treatment mean of each outcome is calculated for event-times $t < 0$, excluding RDOs.

Table B.9: Arrest Type: Long Sample

Dependent Variable:	(1) Any Index Arrests	(2) Any Traffic Arrests	(3) Other Arrests	(4) Any Arrests with No Charge
Treatment $\times$ Post	-0.0010 (0.0019)	-0.0002 (0.0023)	0.0050 (0.0029)	-0.0009 (0.0008)
Pre-Period Mean	0.0197	0.0288	0.0826	0.0035
Observations	1061472	1061472	1061472	1061472

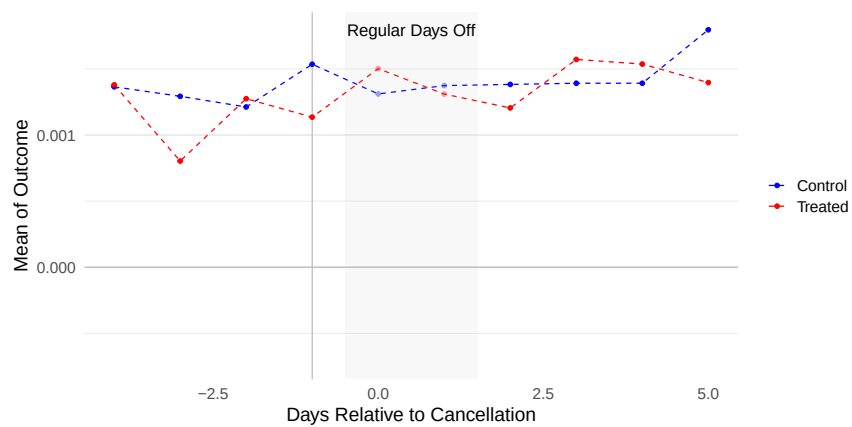
In this specification, we exclude observations falling on RDOs. We additionally include observations from the two work cycles before and after the focal treatment – and omit all control group officers who were treated within these four work cycles. Results are estimated using an aggregated version of our dynamic event-study specification, Equation 1. We report the coefficients on interaction between treatment and a post-treatment indicator, omitting coefficients for event-times $t < 1$. Standard errors clustered by area-DOG are reported in parentheses. The pre-treatment mean of each outcome is calculated for event-times $t < 0$, excluding RDOs.

Table B.10: Pre-Trend Tests: Main Sample & Main Specification

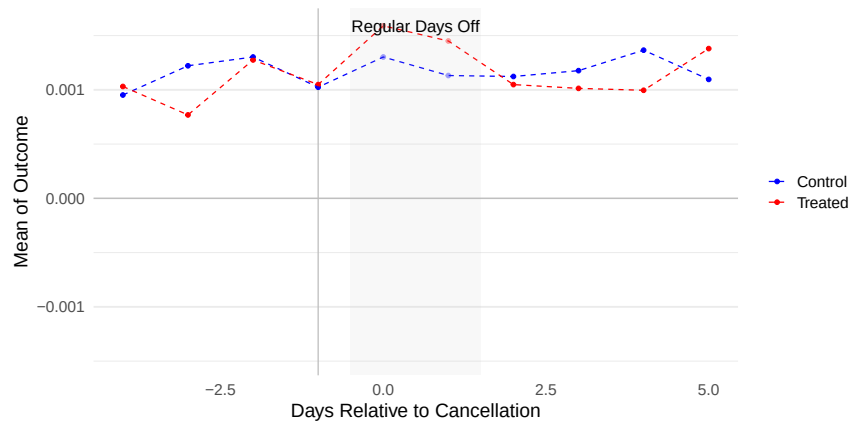
	<i>(A) Attendance and Work</i>		<i>(B) Officer Wellness</i>	
Dependent Variable:	(1) Reports to Work	(2) Total Overtime Hours	(3) Injured on the Job	(4) Calls in Sick
F-stat	302.04	201.06	1.52	2.00
p-value	0.00	0.00	0.23	0.14
Pre-Period Treatment Mean	0.67	0.47	0.05	0.07
Pre-Period Treatment SD	0.47	1.60	0.21	0.25
Pre-Period Control Mean	0.67	0.52	0.04	0.07
Pre-Period Control SD	0.47	1.68	0.20	0.25
Wtd. Correlation	0.51	0.56	0.14	0.55
	<i>(C) Officer Productivity</i>		<i>(D) Enforcement Activity</i>	
Dependent Variable:	(5) Total Calls for Service	(6) Total Investigatory Stops	(7) Any Arrests	(8) Any Use of Force
F-stat	20.26	6.78	3.96	6.35
p-value	0.00	0.00	0.02	0.00
Pre-Period Treatment Mean	3.08	0.08	0.10	0.00
Pre-Period Treatment SD	4.17	0.42	0.30	0.05
Pre-Period Control Mean	3.07	0.08	0.10	0.00
Pre-Period Control SD	4.16	0.42	0.29	0.05
Wtd. Correlation	0.31	0.45	0.43	0.01

C Appendix Figures

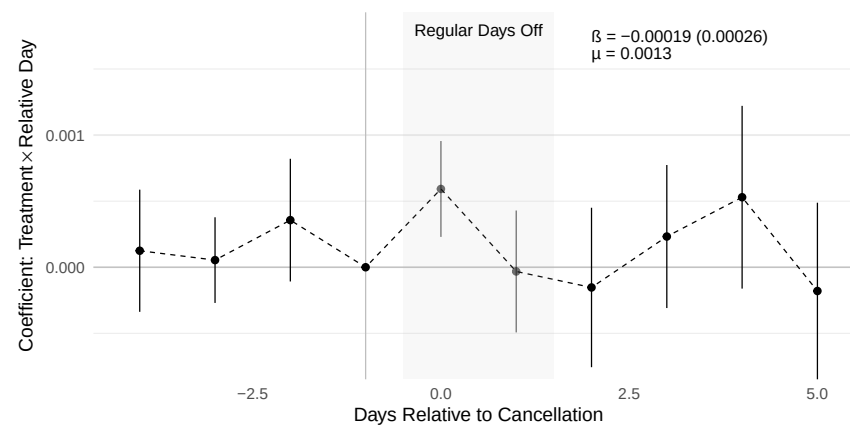
C.1 Compliance with Work Assignments



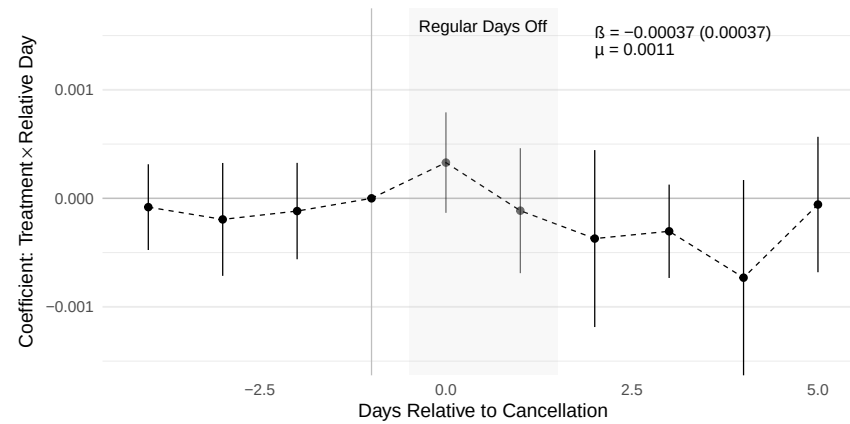
(a) Raw Data: DOG Assignment



(c) Raw Data: Area Assignment



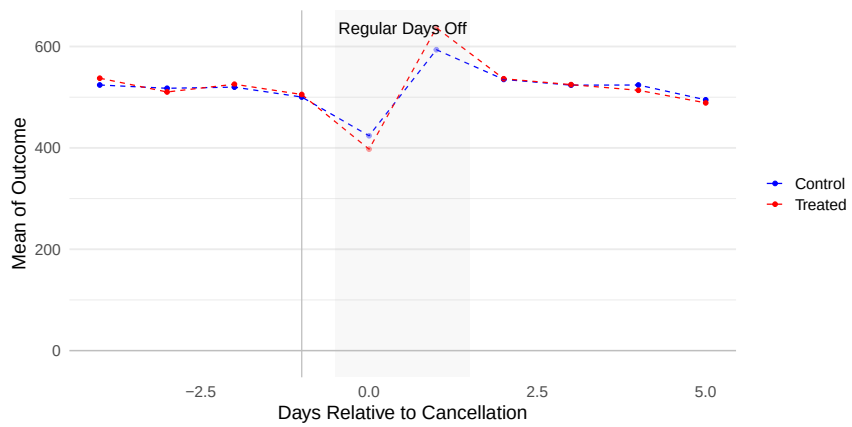
(b) Event Study: DOG Assignment



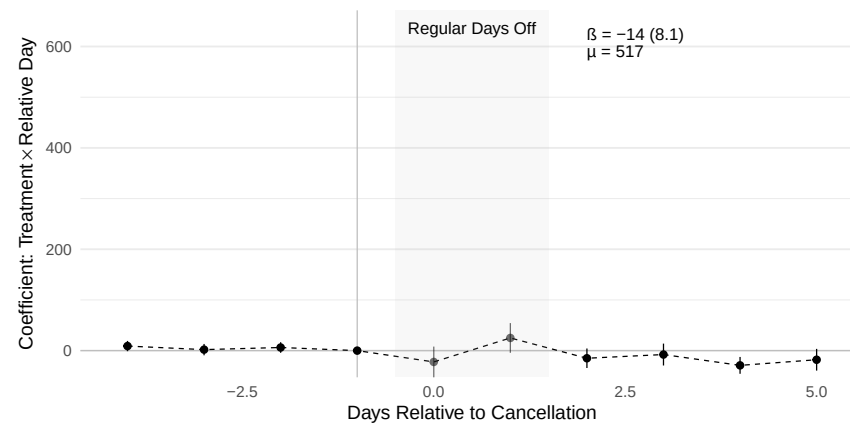
(d) Event Study: Area Assignment

Figure C.9: Change in Work Assignment

C.2 Call Response Time



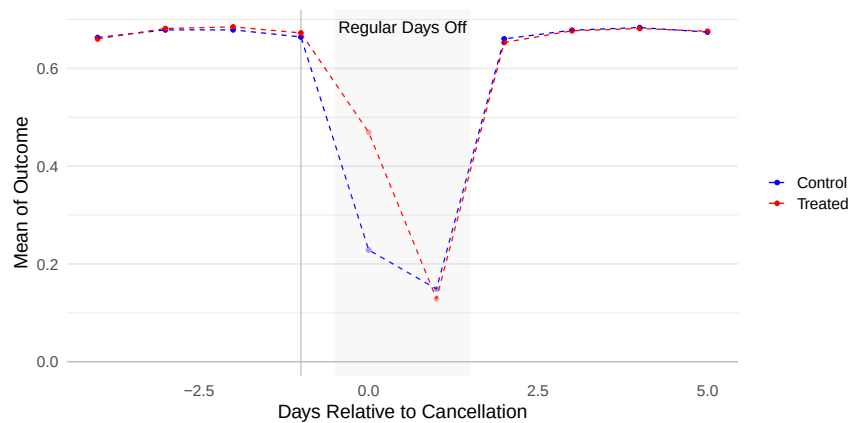
(a) Binscatter Plot



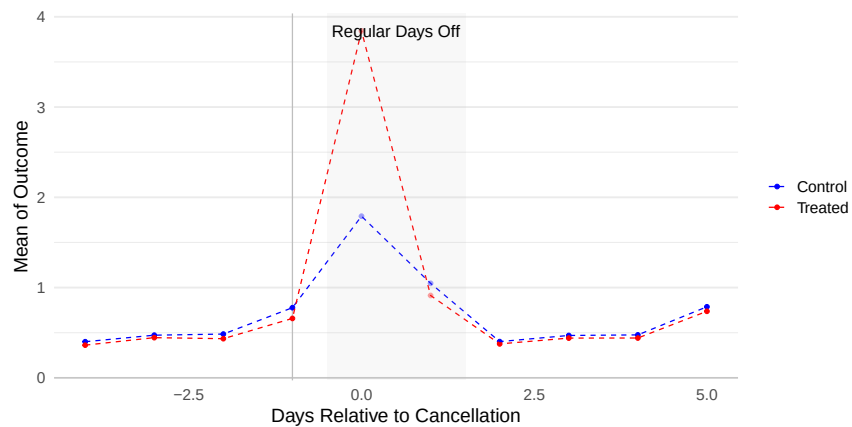
(b) Dynamic DID Event Study

Figure C.10: Time from Call Received by Dispatcher to On-Scene in Seconds

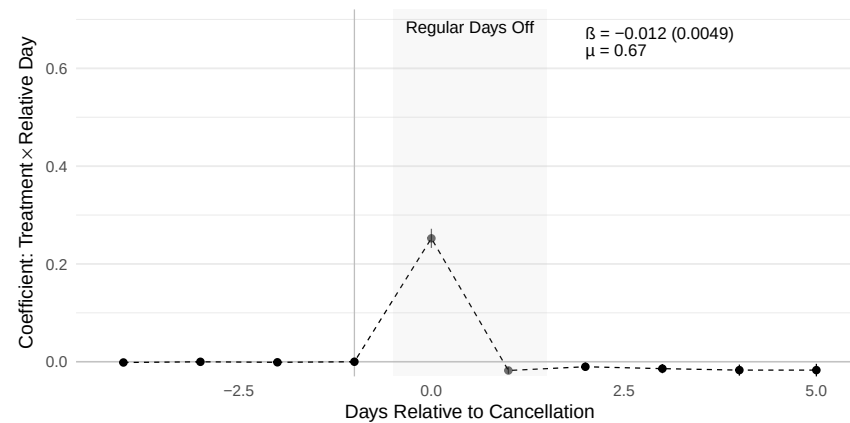
C.3 Event Study Plots: Restricted Sample



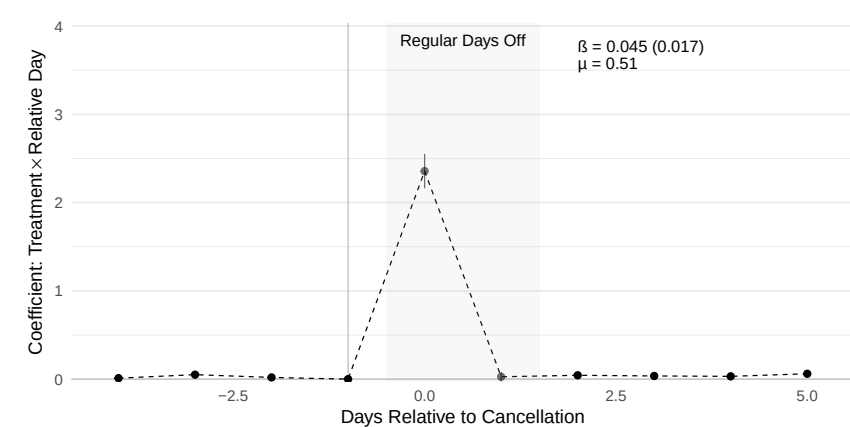
(a) Raw Data: At Work



(c) Raw Data: Overtime Hours

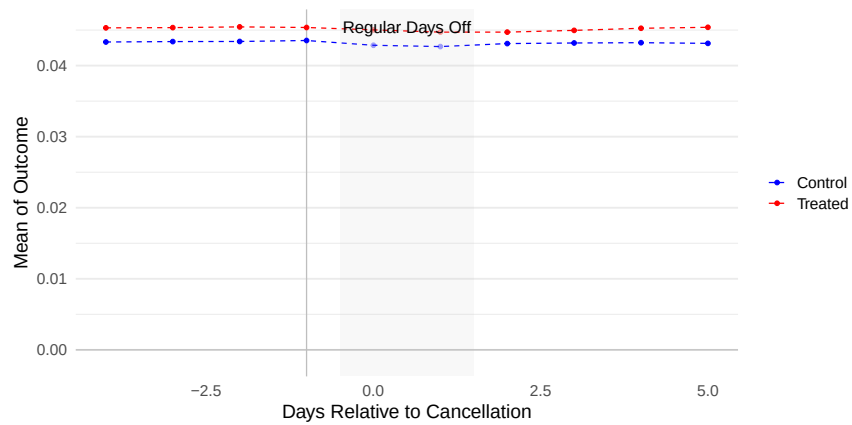


(b) Event Study: At Work

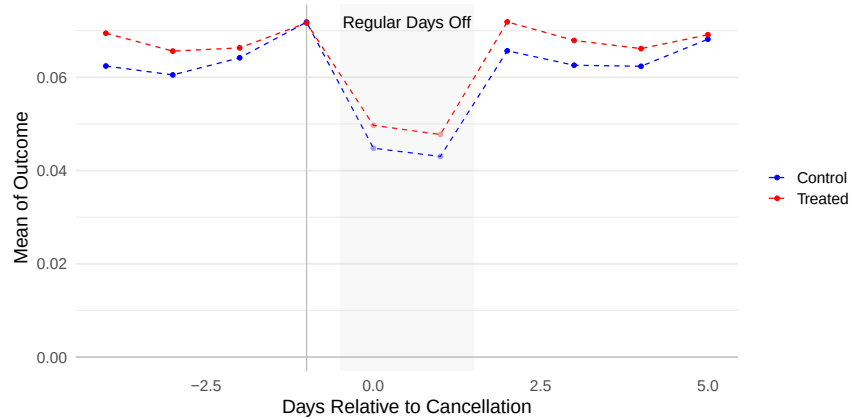


(d) Event Study: Overtime Hours

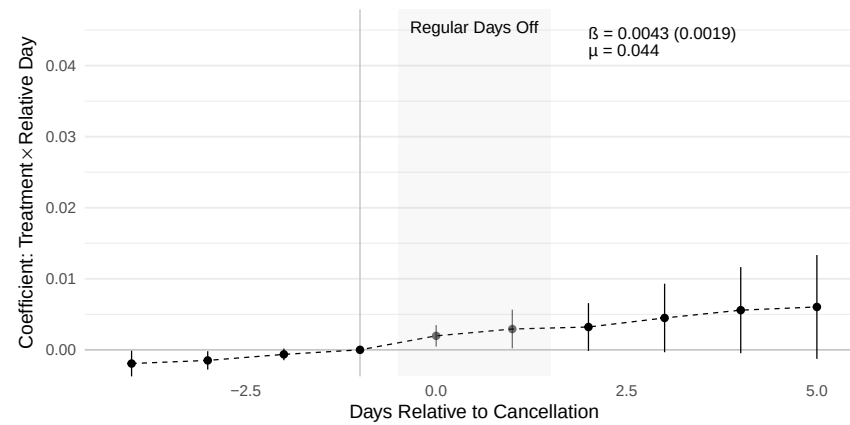
Figure C.11: Treatment Compliance and Effects on Attendance



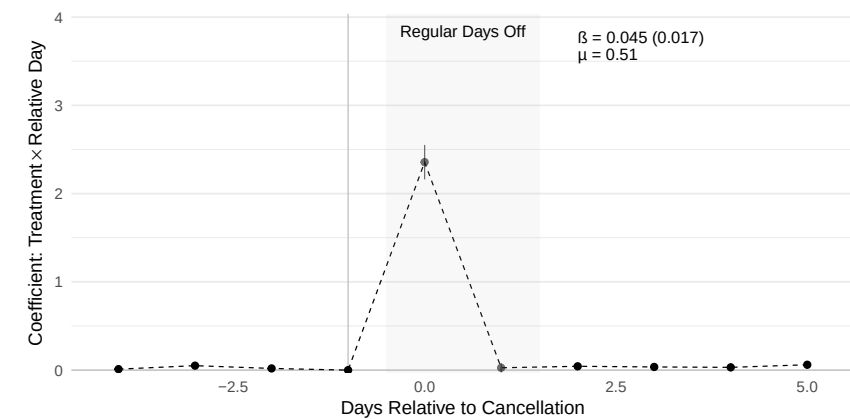
(a) Raw Data: Injury



(c) Raw Data: Calls in Sick

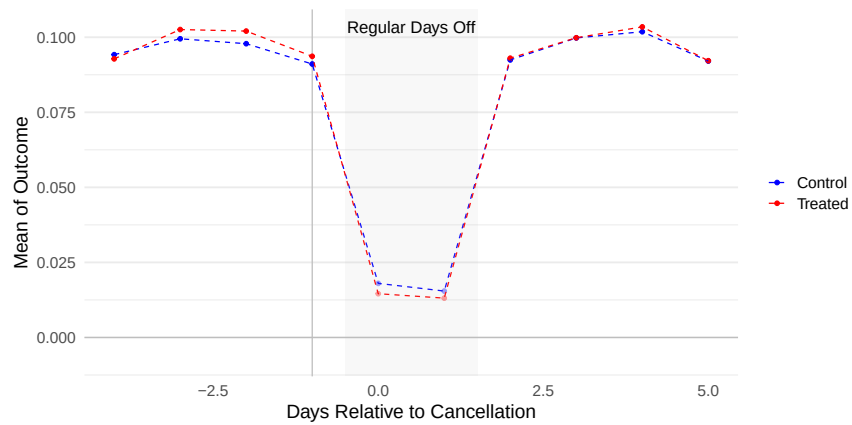


(b) Event Study: Injury

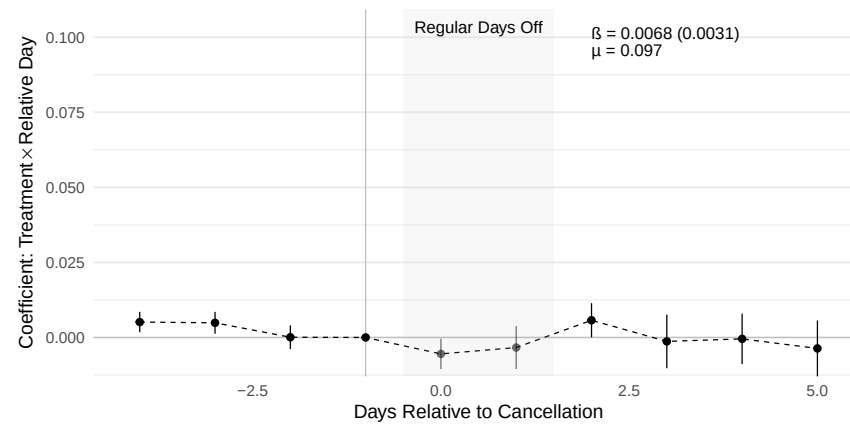


(d) Event Study: Calls in Sick

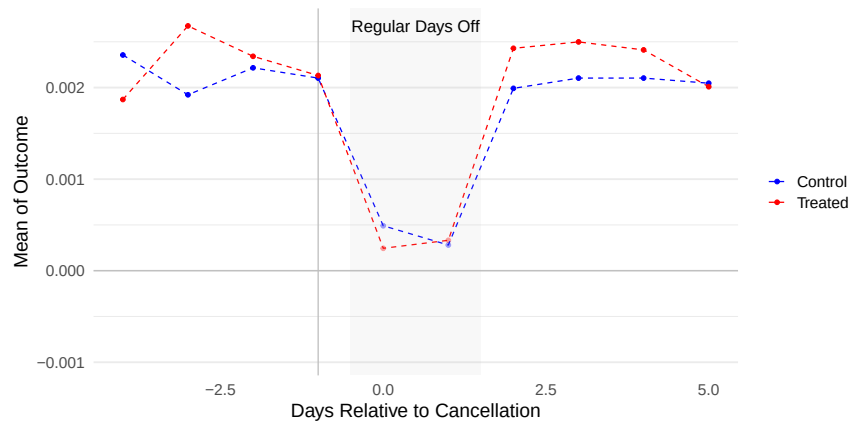
Figure C.12: Effects on Officer Wellness



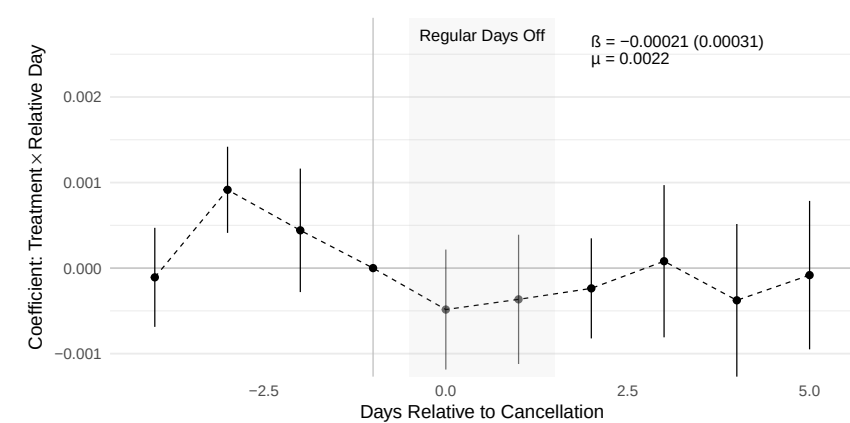
(a) Raw Data: Arrests



(b) Event Study: Arrests

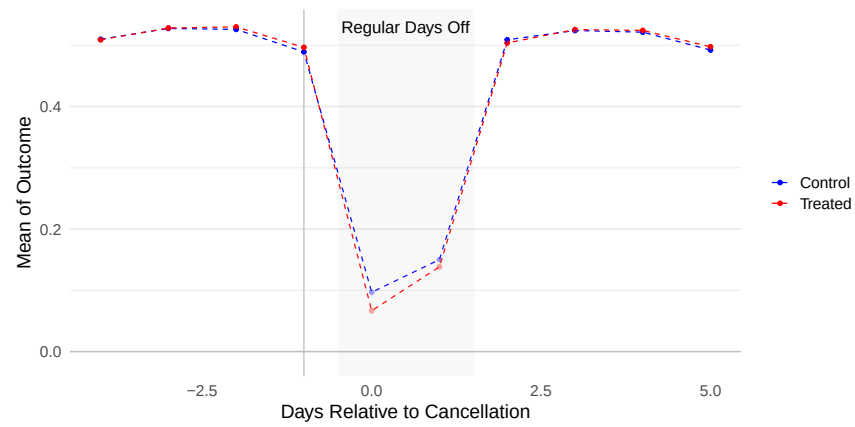


(c) Raw Data: Use of Force

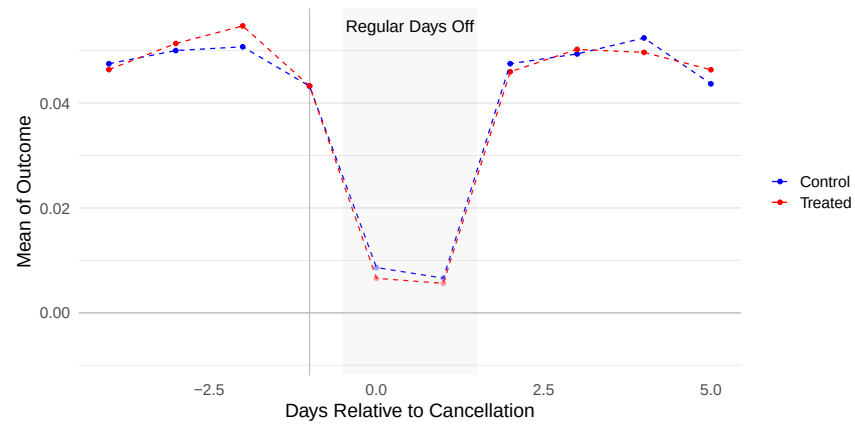


(d) Event Study: Use of Force

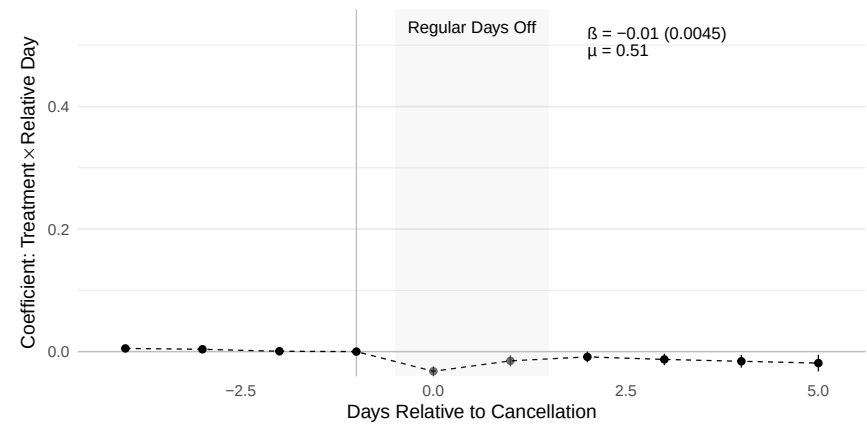
Figure C.13: Effects on Law Enforcement Activity



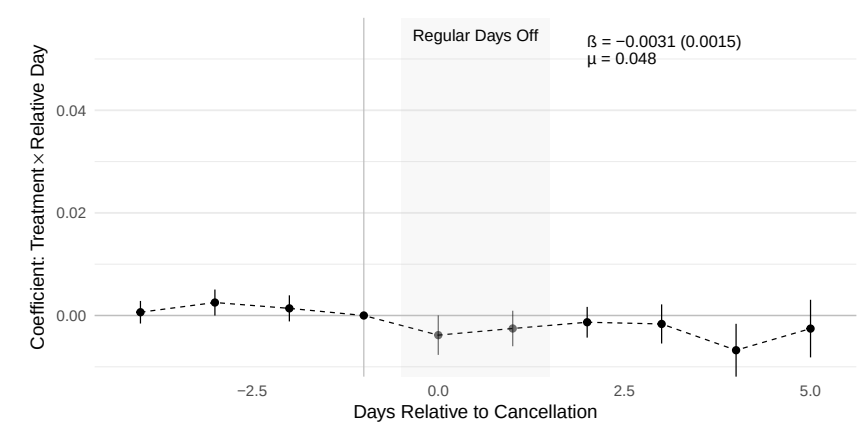
(a) Raw Data: Calls for Service



(c) Raw Data: Stops



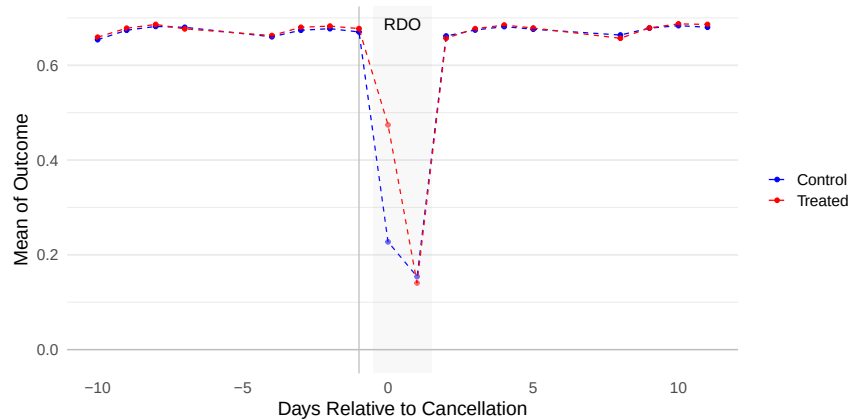
(b) Event Study: Calls for Service



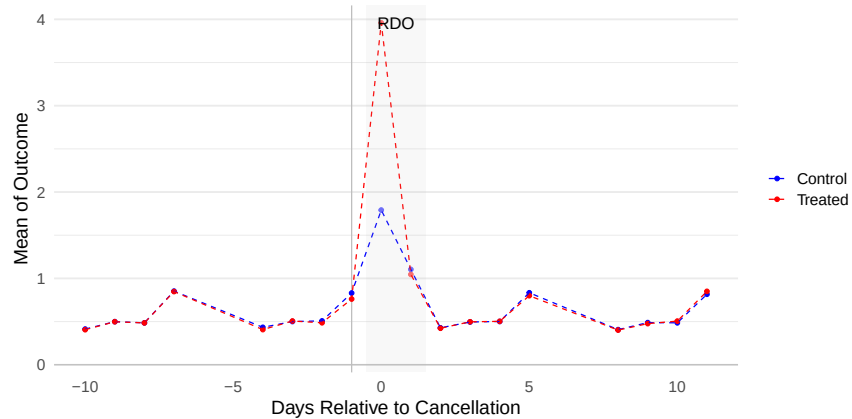
(d) Event Study: Stops

Figure C.14: Effects on Productivity

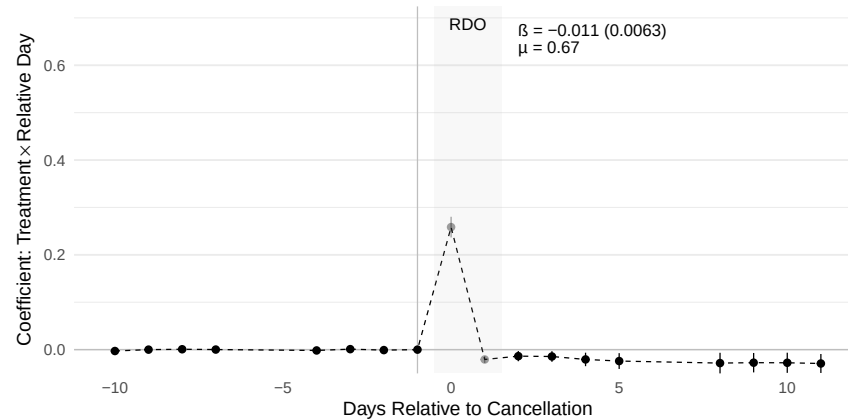
C.4 Event Study Plots: Long Sample



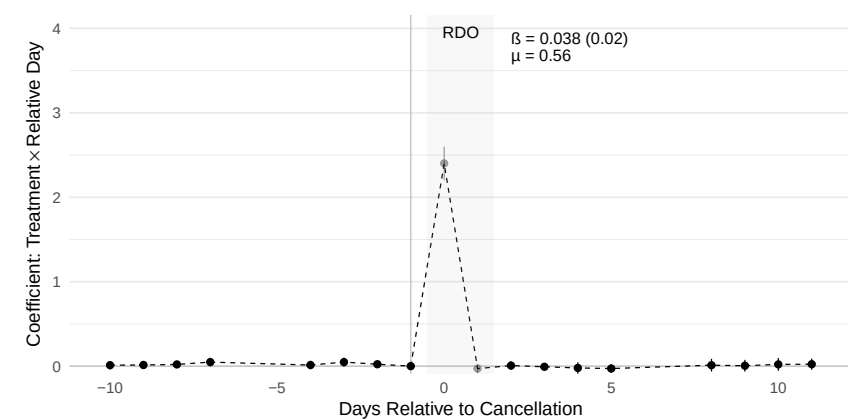
(a) Raw Data: At Work



(c) Raw Data: Overtime Hours

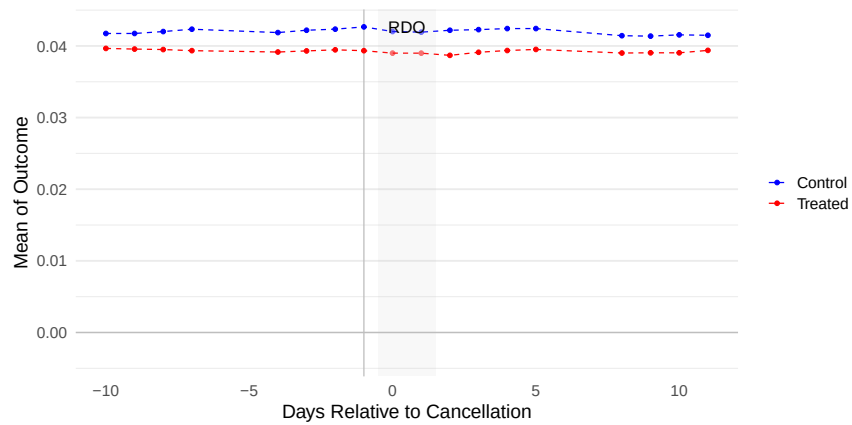


(b) Event Study: At Work

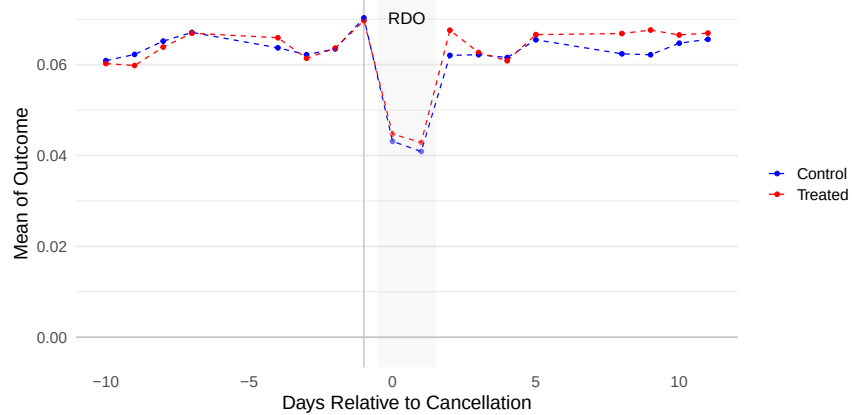


(d) Event Study: Overtime Hours

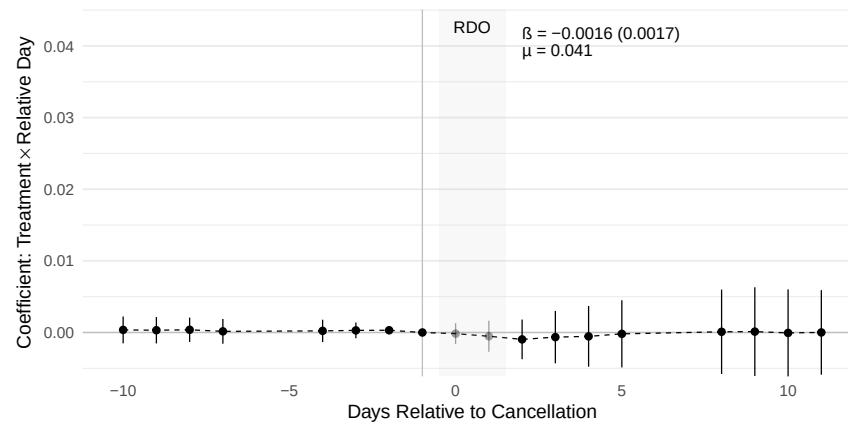
Figure C.15: Treatment Compliance and Effects on Attendance



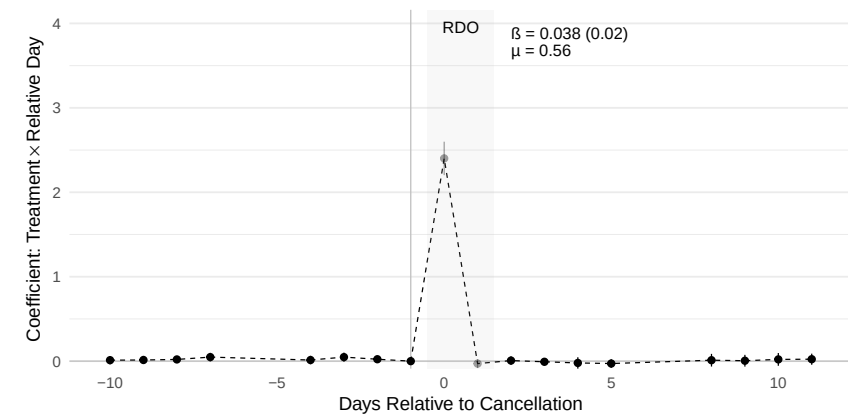
(a) Raw Data: Injury



(c) Raw Data: Calls in Sick

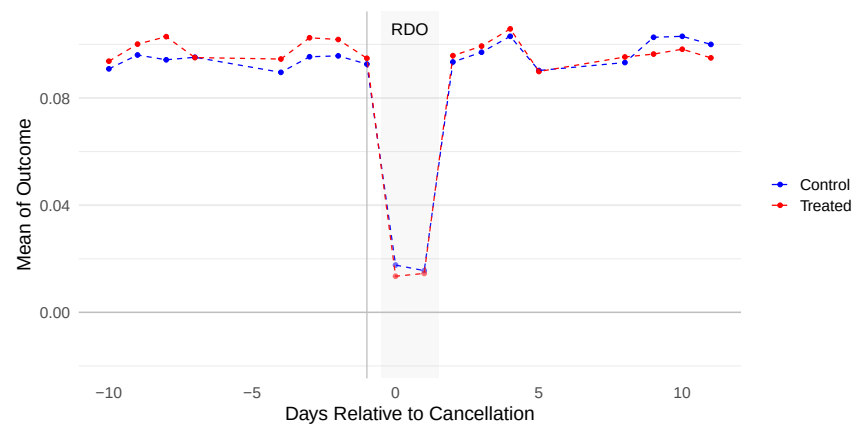


(b) Event Study: Injury

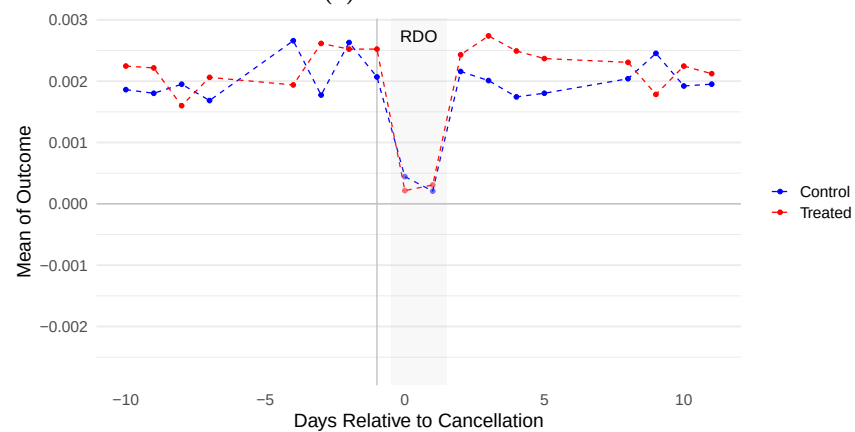


(d) Event Study: Calls in Sick

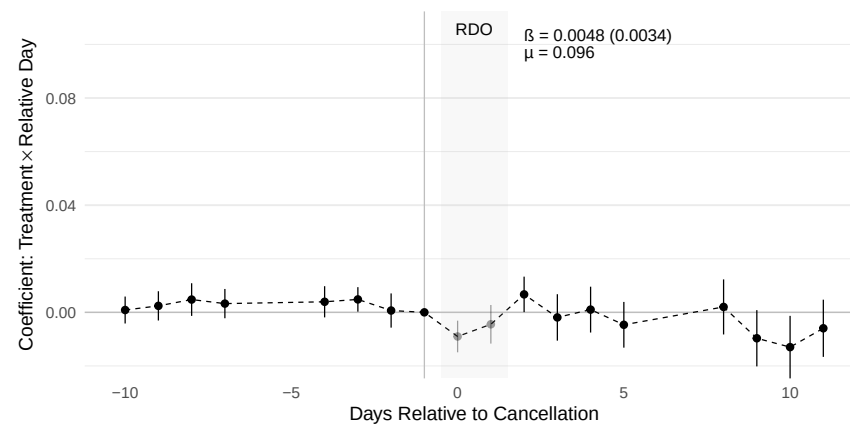
Figure C.16: Effects on Officer Wellness



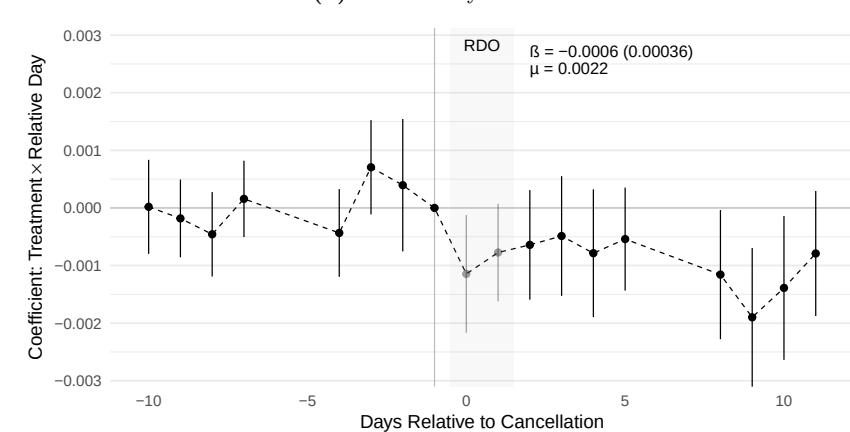
(a) Raw Data: Arrests



(c) Raw Data: Use of Force

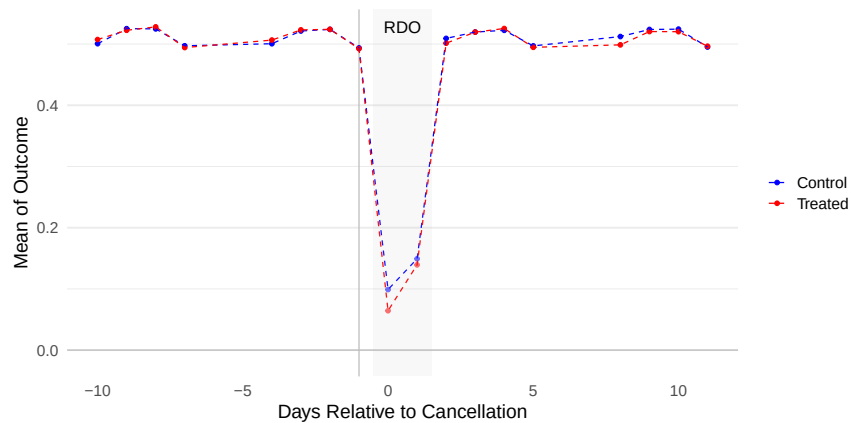


(b) Event Study: Arrests

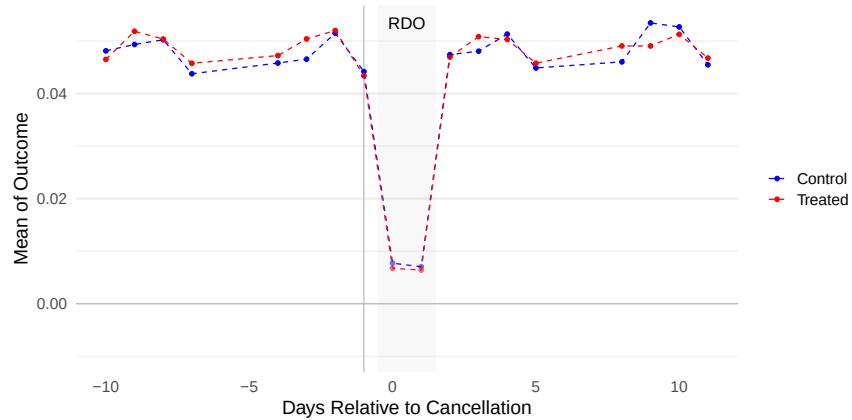


(d) Event Study: Use of Force

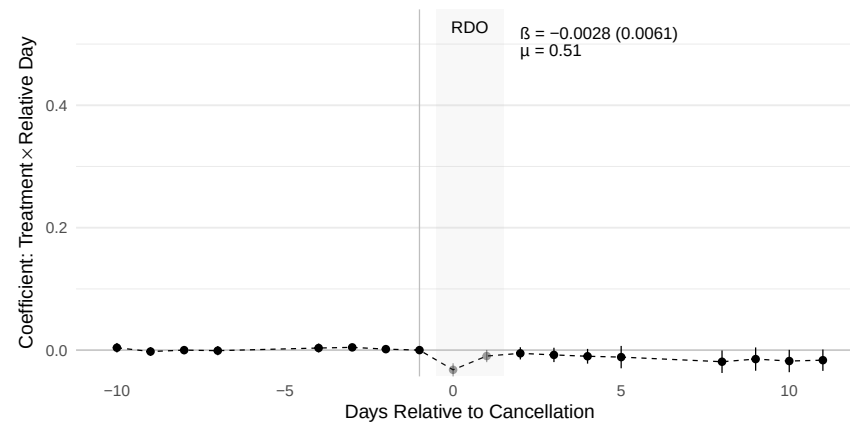
Figure C.17: Effects on Law Enforcement Activity



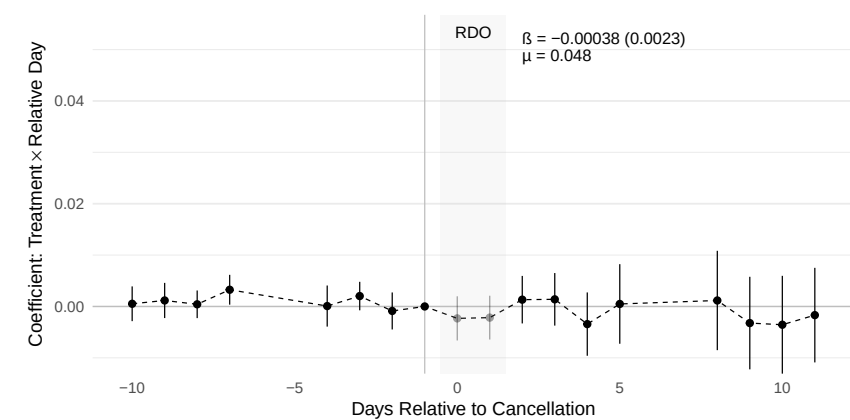
(a) Raw Data: Calls for Service



(c) Raw Data: Stops



(b) Event Study: Calls for Service



(d) Event Study: Stops

Figure C.18: Effects on Productivity